



TUESDAY, JUNE 23, 2020

IEC 61439 & IEC60947 Standards

Standard for LV Switchboards & LV Circuit Breakers

Product Marketing – Electrification business – Smart Power, United Arab Emirates



Introduction to IEC 61439 Standard

Switchboard Standard

Present situation

The IEC 61439-1 is a general rules and contains characteristics, properties and performance common to all the switchboards

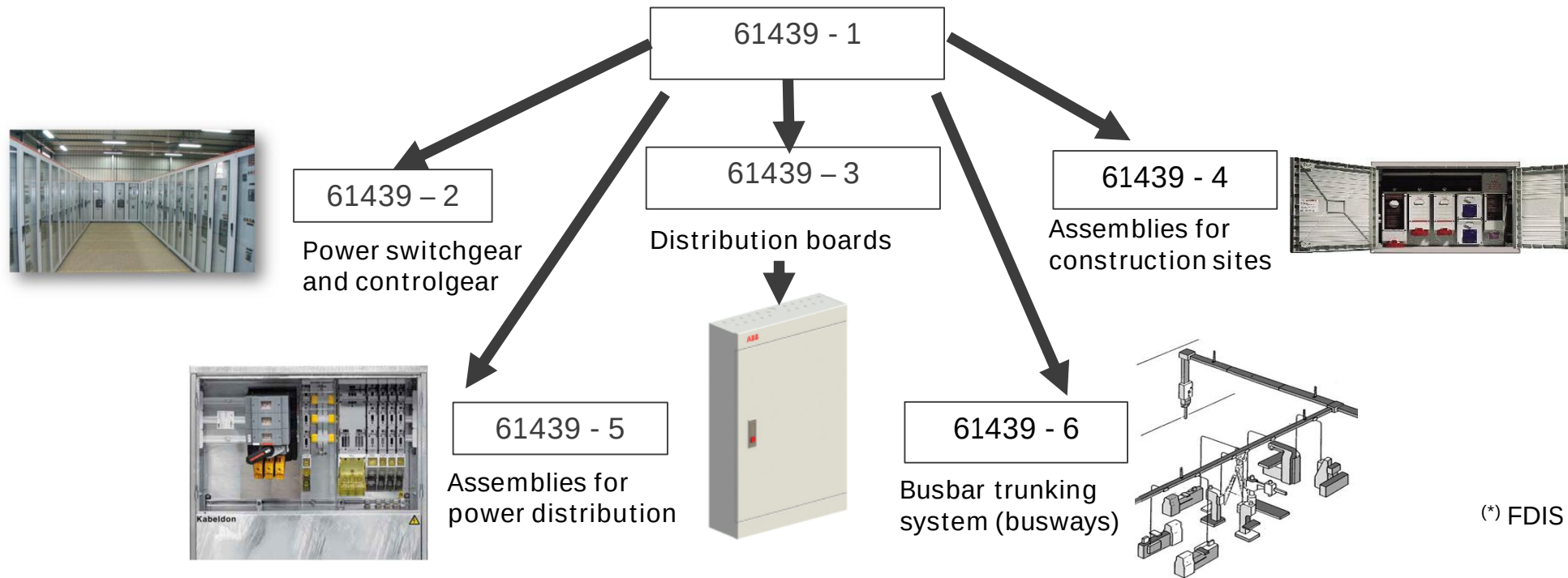


ABB Guidelines for IEC 61439 TTA program - Key Inputs for testing

Product certification – Switchgear & controlgear assemblies

Design verification as per IEC 61439-1:
Construction

- 1) Strength of materials & parts
- 2) Degree of protection of enclosures
- 3) Clearance & creepage distances
- 4) Protection against electric shock and integrity of protective circuits
- 5) Incorporation of switching devices and components
- 6) Internal electrical circuits and connections
- 7) Terminals for external conductors

Performance

- 8) Dielectric properties
- 9) Verification of temperature rise
- 10) Short circuit withstand strength
- 11) Electromagnetic compatibility
- 12) Mechanical operation



Construction & performance features-General Rules

Scope:

Assemblies intended / designed

- a) Rated voltage does not exceed 1 000 V AC or 1500 V DC.
- b) Stationary or movable assemblies with or without encl.
- c) Generation, transmission, distribution and control of electric energy.
- d) Special service conditions, for example in ships and in rail vehicles provided that the other relevant specific requirements are complied with.
- e) Electrical equipment of machines provided that the other relevant specific requirements are complied with.

Service conditions:

- 1) Ambient air temperature:
 - 5°C to +40°C for indoor installation
 - 25°C to +40°C for outdoor installation
- 2) Humidity:
 - ≤50% at max. 40°C, higher humidity is permitted at lower temperatures for indoor installations
 - ≤100% at max. 25°C for outdoor
- 3) Pollution degree: (1, 2, 3 or 4)
- 4) Altitude: shall not exceed 2000m.
- 5) Special service conditions like dust, temperature, humidity, magnetic fields, smoke, fungus etc.
(special agreements between manufacturer & user)

IEC 61439 TTA design validation –Construction & performance features

Enlisted tests as specified by IEC 61439 1-2 :

Sl.No.	Test to be Verified	Clause No.	Verification options available		
			Testing	Comparison with a reference design	Assessment
1	Strength of material and parts	10.2			
	Resistance to corrosion	10.2.2	Yes	No	No
	Properties of insulating materials	10.2.3	Yes	No	No
	Thermal stability	10.2.3.1	Yes	No	Yes
	Resistance to abnormal heat and fire due to internal electric effects	10.2.3.2	Yes	No	Yes
	Resistance to ultra-violet (UV) radiation	10.2.4	Yes	No	No
	Lifting	10.2.5	Yes	No	No
	Mechanical Impact	10.2.6	Yes	No	No
	Marking	10.2.7	Yes	No	No
2	Degree of protection of enclosures	10.3	Yes	No	Yes
3	Clearances and creepage distances	10.4	Yes	No	No
4	Protection against electric shock and integrity of protective circuits:	10.5			
	Effective continuity between the exposed conductive parts of the ASSEMBLY and the protective circuit	10.5.2	Yes	No	No
	Short-circuit withstand strength of the protective circuit	10.5.3	Yes	Yes	No
5	Incorporation of switching devices and components	10.6	No	No	Yes
6	Internal electrical circuits and connections	10.7	No	No	Yes
7	Terminals for external conductors	10.8	No	No	Yes
8	Dielectric properties:	10.9			
	Power-frequency withstand voltage	10.9.2	Yes	No	No
	Impulse withstand voltage	10.9.3	Yes	No	Yes
9	Temperature-rise limits	10.1	Yes	Yes	Yes
10	Short-circuit withstand strength	10.11	Yes	Yes	No
11	Electromagnetic compatibility (EMC)	10.12	Yes	No	Yes
12	Mechanical Operation	10.13	Yes	No	No

Product certification – Switchgear & controlgear assemblies

Construction requirement tests

1) Strength of materials & parts - Clause: 10.2

Resistance to corrosion - Clause: 10.2.2

Severity test A (for metallic indoor enclosures)

- 6 cycles of 24 h each to damp heat cycling test according to IEC 60068-2-30 (Test Db) at $(40 \pm 3) ^\circ\text{C}$ and relative humidity of 95 %
- 2 cycles of 24 h each to salt mist test according to IEC 60068-2-11; (Test Ka: Salt mist), at a temperature of $(35 \pm 2) ^\circ\text{C}$

Severity test B (for metallic outdoor enclosures)

- 5 cycles of 24 h each to damp heat cycling test according to IEC 60068-2-30 (Test Db) at $(40 \pm 3) ^\circ\text{C}$ and relative humidity of 95 %
- 7 cycles of 24 h each to salt mist test according to IEC 60068-2-11; (Test Ka: Salt mist), at a temperature of $(35 \pm 2) ^\circ\text{C}$.

Results to be obtained:

No iron oxide, cracking
Degree of rusting Ri1
Mechanical integrity is intact

Product certification – Switchgear & controlgear assemblies

Construction requirement tests

1) Strength of materials & parts - Clause: 10.2

Properties of insulating materials - Clause: 10.2.3

Results to be obtained:

- No cracks, material have become sticky or greasy
- No traces of the cloth shall remain on the sample
- material of the enclosure or sample shall not stick to the cloth

Verification of thermal stability of enclosures

- The test shall be carried out according to IEC 60068-2-2 Test Bb, at a temperature of 70 °C, with natural air circulation, for a duration of 168 h (7 days) and with a recovery of 96 h (4 days).
- With the forefinger wrapped in a dry piece of rough cloth, the sample is pressed with a force of 5 N.

Verification of resistance of insulating materials to abnormal heat and fire due to internal electric effects (As per IEC 60695)

Glow wire test

- 960 °C for parts necessary to retain current-carrying parts in position (busbar supports)
- 850 °C for enclosures intended for mounting in hollow walls
- 650 °C for all other parts, including parts necessary to retain the protective conductor

Product certification – Switchgear & controlgear assemblies

Construction requirement tests

1) Strength of materials & parts - Clause: 10.2

Resistance to ultra-violet (UV) radiation - Clause: 10.2.4

– The UV radiation causes deterioration of synthetic material use for enclosures.

Procedure:

UV Test according to ISO 4892 – 2 method A; 1000 cycles of 5 min of watering and 25 min. of dry period with xenon lamp providing a total test period of 500 hrs.

The value of temperature and humidity for the test are $(65 \pm 3)^\circ\text{C}$ and $(65 \pm 5) \%$.

NOTE: This test is applicable only for enclosures and external parts of ASSEMBLIES intended to be installed outdoors.

Results to be obtained:

- Samples shall not show cracks or deterioration visible to normal or corrected vision without any magnification

Product certification – Switchgear & controlgear assemblies

Construction requirement tests

1) Strength of materials & parts - Clause: 10.2

Results to be obtained:

- After the test, with the test weights in place, the ASSEMBLY shall show no cracks or permanent distortions visible to normal or corrected vision

Lifting - Clause: 10.2.5

- From a standstill position, the ASSEMBLY shall be raised smoothly without jerking in a vertical plane to a height of ≥ 1 m and lowered in the same manner to a standstill position
- This test is repeated a further two times after which the ASSEMBLY is raised up and suspended clear of the floor for 30 min without any movement.
- Following this test the ASSEMBLY shall be raised smoothly without jerking from a standstill position to a height of ≥ 1 m and moved $(10 \pm 0,5)$ m horizontally, then lowered to a standstill position
- This sequence, shall be carried out three times at uniform speed, each sequence being carried out within 1 min.

Product certification – Switchgear & controlgear assemblies

Construction requirement tests

Mechanical Impact - Clause: 10.2.6

Mechanical impact tests where required by the specific ASSEMBLY standard are to be carried out in accordance with IEC 62262

Procedure:

- Number of impacts on each exposed face shall be five unless otherwise specified in relevant product standard.
- The impacts shall be evenly distributed on the faces of the enclosure under test.
- At the surrounding of any point of enclosure, not more than three impacts shall be applied.

Results to be obtained:

Structural integrity shall be maintained.
Degree of protection (IP) shall be unaffected.
Functionality of electrical products shall not be hampered.

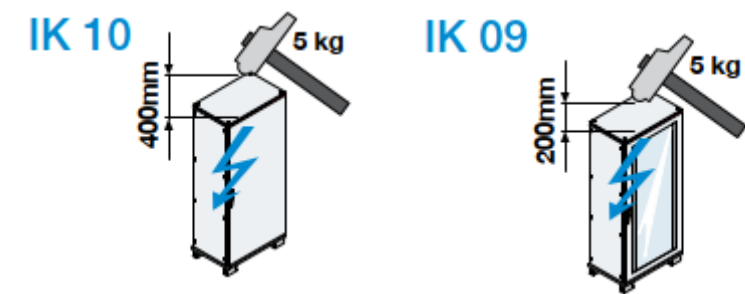


Table 1 – Relation between IK code and impact energy

IK code	IK00	IK01	IK02	IK03	IK04	IK05	IK06	IK07	IK08	IK09	IK10
Impact energy, J	*	0,14	0,2	0,35	0,5	0,7	1	2	5	10	20
* Not protected according to this standard.											
NOTE 1 When higher impact energy is required, the value of 50 J is recommended.											
NOTE 2 A characteristic group numeral of two figures has been chosen to avoid confusion with some national standards which used a single numeral for a specific impact energy.											

Product certification – Switchgear & controlgear assemblies

Construction requirement tests

1) Strength of materials & parts - Clause: 10:2

Results to be obtained:

- After the test the marking shall be legible to normal or corrected vision

Marking - Clause: 10.2.7

- Marking made by molding, pressing, engraving or similar, including labels with a laminated plastic covering, shall not be submitted to the following test
- The test is made by rubbing the marking by hand for 15 s with a piece of cloth soaked in water and then for 15 s with a piece of cloth soaked with petroleum spirit

Product certification – Switchgear & controlgear assemblies

2) Degree of protection of enclosures - Clause: 10.3

IP degrees of protection

1st NUMBER: protection against solid bodies

0 No protection

1 Protected against solid bodies larger than 50mm (accidental contact with hands)

2 Protected against solid bodies larger than 12 mm (fingers)

3 Protected against solid bodies larger than 2.5mm (tools, wires)

4 Protected against solid bodies larger than 1 mm

5 Protected against dust (no harmful deposit)

6 Totally protected against dust

2nd NUMBER: protection against liquids

0 No protection

1 Protected against vertical fall of drops of water (condensation)

2 Protected against drops of water up to 15° from vertical

3 Protected against drops of water up to 60° from vertical

4 Protected against splashes of water from all directions

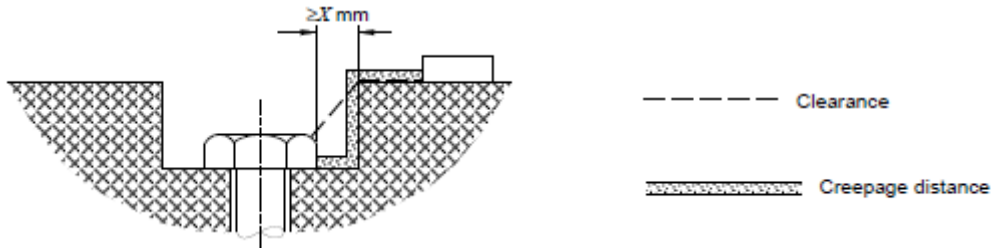
5 Protected against jets of water from all directions by means of a hose

6 Protection against jets of water similar to sea waves

As per IEC 60529 ArTu K is rated to IP43/IP54/IP65

Product certification – Switchgear & controlgear assemblies

3) Clearances and creepage distances – Clause:10.4



Clearance - the distance between two conductive parts along a string stretched the shortest way between these conductive parts.

Table 1 – Minimum clearances in air ^a (8.3.2)

Rated impulse withstand voltage U_{imp} kV	Minimum clearance mm
$\leq 2,5$	1,5
4,0	3,0
6,0	5,5
8,0	8,0
12,0	14,0

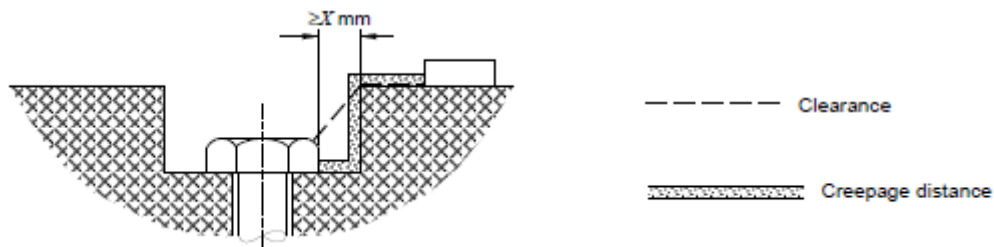
^a Based on inhomogeneous field conditions and pollution degree 3.

➡ Clearance (as per Table -1):

- ➡ Based on 12kV U_{imp}
Main Busbar and Incoming unit (Clearance ≥ 14 mm)
- ➡ Based on 8kV U_{imp}
Outgoing Circuit feeders (Clearance ≥ 8 mm)
- ➡ Based on 6kV U_{imp}
Outgoing Circuit starters (Clearance ≥ 5.5 mm)

Product certification – Switchgear & controlgear assemblies

3) Clearances and creepage distances – Clause:10.4



Creepage distance - the shortest distance along the surface of a solid insulating material between two conductive parts.

- ➡ Creepage (Pollution Degree 3, material group IIIa) :
 - Based on U_i 1000V
 - ➡ Main Busbar and Incoming unit (Clearance ≥ 16 mm)
 - Based on U_i 800V
 - ➡ Outgoing Circuit feeders (Clearance ≥ 12.5 mm)
 - Based on U_i 600V
 - ➡ Outgoing Circuit starters (Clearance ≥ 10 mm)

Table 2 – Minimum creepage distances (8.3.3)

Rated insulation voltage U_i	Minimum creepage distance mm							
	Pollution degree							
	1	2			3			
	Material group ^c	Material group ^c			Material group ^c			
V^b	All material groups	I	II	IIIa and IIIb	I	II	IIIa	IIIb
32	1,5	1,5	1,5	1,5	1,5	1,5	1,5	1,5
40	1,5	1,5	1,5	1,5	1,5	1,6	1,8	1,8
50	1,5	1,5	1,5	1,5	1,5	1,7	1,9	1,9
63	1,5	1,5	1,5	1,5	1,6	1,8	2	2
80	1,5	1,5	1,5	1,5	1,7	1,9	2,1	2,1
100	1,5	1,5	1,5	1,5	1,8	2	2,2	2,2
125	1,5	1,5	1,5	1,5	1,9	2,1	2,4	2,4
160	1,5	1,5	1,5	1,6	2	2,2	2,5	2,5
200	1,5	1,5	1,5	2	2,5	2,8	3,2	3,2
250	1,5	1,5	1,8	2,5	3,2	3,6	4	4
320	1,5	1,6	2,2	3,2	4	4,5	5	5
400	1,5	2	2,8	4	5	5,6	6,3	6,3
500	1,5	2,5	3,6	5	6,3	7,1	8,0	8,0
630	1,8	3,2	4,5	6,3	8	9	10	10
800	2,4	4	5,6	8	10	11	12,5	a
1 000	3,2	5	7,1	10	12,5	14	16	
1 250	4,2	6,3	9	12,5	16	18	20	
1 600	5,6	8	11	16	20	22	25	

NOTE 1 The CTI values refer to the values obtained in accordance with IEC 60112:2003, method A, for the insulating material used.

NOTE 2 Values taken from IEC 60664-1, but maintaining a minimum value of 1,5 mm.

a Insulation of material group IIIb is not recommended for use in pollution degree 3 above 630 V.

b As an exception, for rated insulation voltages 127, 208, 415, 440, 660/690 and 830 V, creepage distances corresponding to the lower values 125, 200, 400, 630 and 800 V may be used.

c Material groups are classified as follows, according to the range of values of the comparative tracking index (CTI) (see 3.6.16):

- Material group I 600 $\leq$ CTI
- Material group II 400 $\leq$ CTI < 600
- Material group IIIa 175 $\leq$ CTI < 400
- Material group IIIb 100 $\leq$ CTI < 175

Product certification – Switchgear & controlgear assemblies

4) Protection against electric shock and integrity of protective circuits - Clause: 10.5

- Effective earth continuity between the exposed conductive parts of the ASSEMBLY and the protective circuit - Clause: 10.5.2
 - exposed conductive parts of the ASSEMBLY are effectively connected to the terminal for the incoming external protective conductor and that the resistance of the circuit does not exceed $0,1 \Omega$
 - Short-circuit withstand strength of the protective circuit : Clause: 10.5.3
- ➡ Effectiveness of the assembly for external faults (10.5.3):

Icw 60kA/1sec (6300A, 5500A & 5000A)

Icw 39kA/1sec (4000A & 3200A)

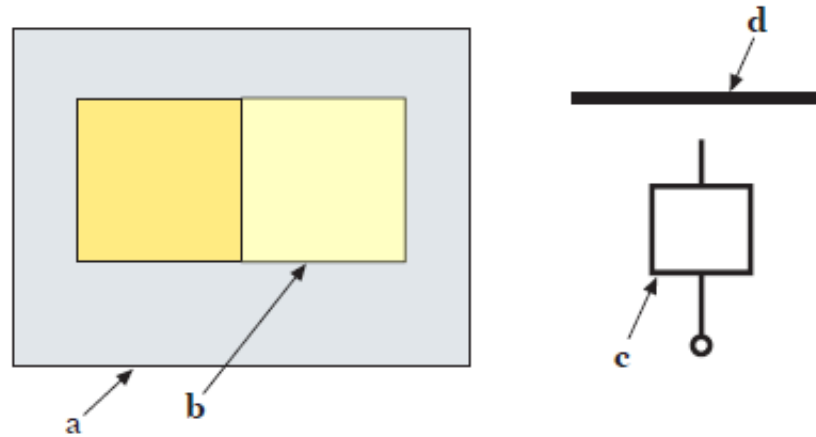
Icw 30kA/1sec (2500A, 1600A & 1250A)

Icw 21.6kA/1sec (800A & 400A)

Product certification – Switchgear & controlgear assemblies

Segregation Forms:

Symbolic representation



a - Housing

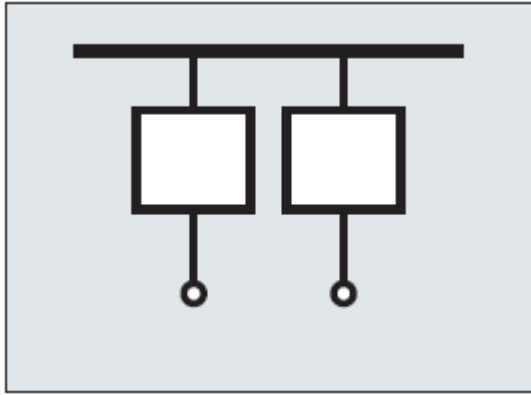
b – Internal segregation

c – Functional units including terminals for external conductors

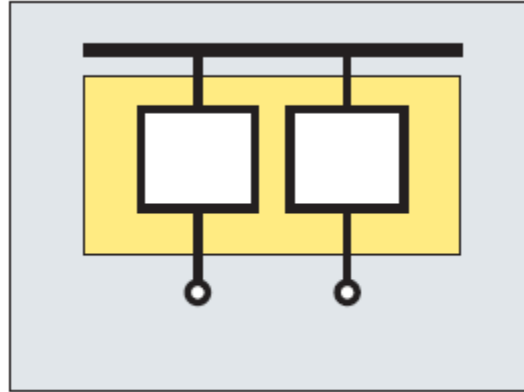
d – Busbars, main & distribution.

Product certification – Switchgear & controlgear assemblies

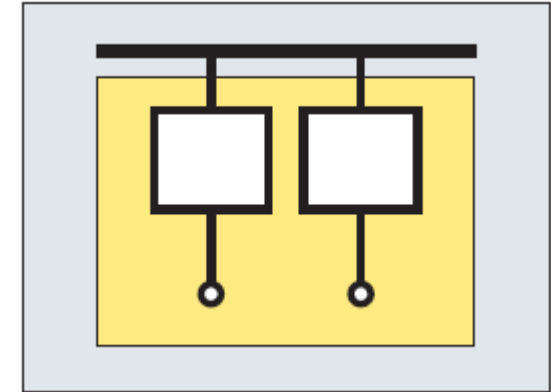
Segregation Forms:



Form 1
no internal segregation



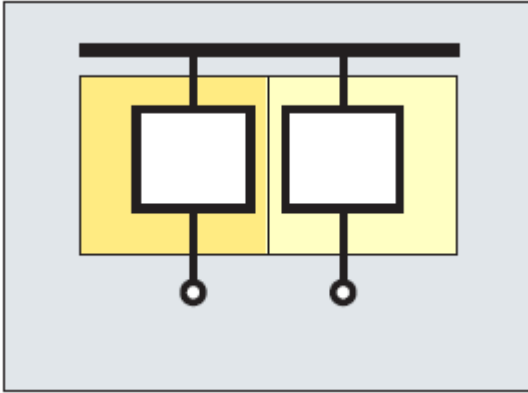
Form 2
segregation of busbar from
functional units
Form 2a
Terminals not separated from the
busbars



Form 2b
Terminals separated from
busbars

Product certification – Switchgear & controlgear assemblies

Segregation Forms:

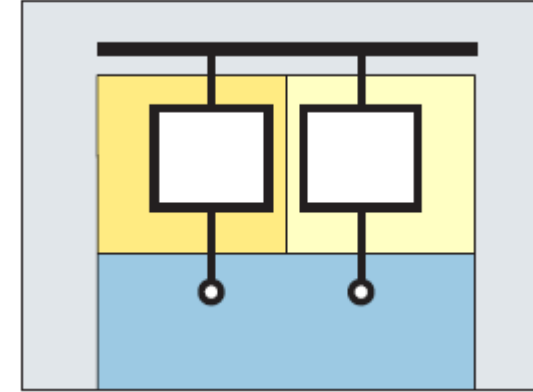


Form 3

separation of the busbars from the functional units + separation of the functional units from each other

Form 3a

Terminals not separated from the busbars

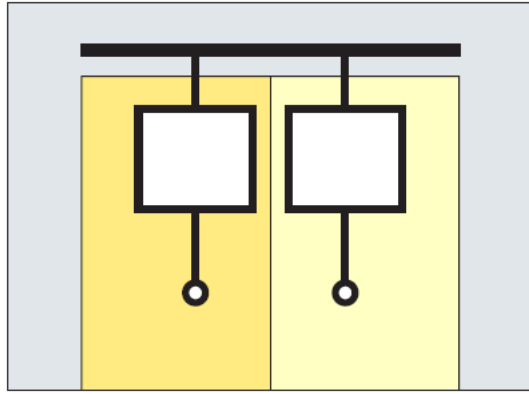


Form 3b

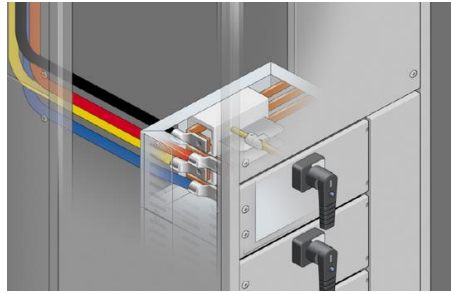
Terminals separated from the busbars.

Product certification – Switchgear & controlgear assemblies

Segregation Forms:



Form 4a Type 2 (not possible in ArTu K)

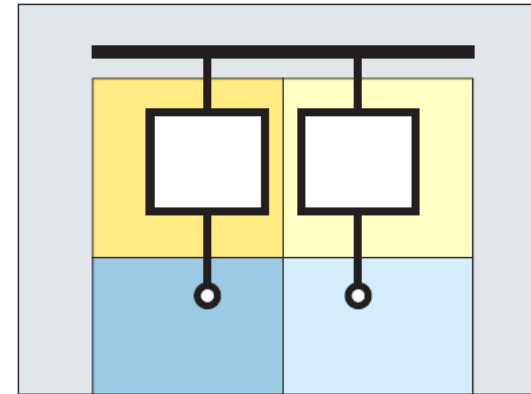


Form 4

separation of the busbars from the functional units
+ separation of the functional units from each other
+ separation of the terminals from each other

Form 4a

Terminals in the same compartment as the associated functional unit



Form 4b Type 6



Form 4b Type 7



Form 4b

Terminals not in the same compartment as the associated functional unit

Product certification – Switchgear & controlgear assemblies

5) Incorporation of switching devices and components - Clause:10.6

Selection of switching devices and components

Installation of switching devices and components

Accessibility

Barriers

Direction of operation and indication of switching positions

Indicator lights and push-buttons

6) Internal electrical circuits and connections- Clause 10.7

Main circuits

Auxiliary circuits

7) Terminals for external conductors - Clause:10.8

Product certification – Switchgear & controlgear assemblies

Performance

8) Dielectric Properties - Clause:10.9

Power frequency withstand voltage: Clause:10.9.2

Table 8 – Power-frequency withstand voltage for main circuits (10.9.2)

Rated insulation voltage U_i (line to line a.c. or d.c.)	Dielectric test voltage a.c. r.m.s.	Dielectric test voltage ^b d.c.
V	V	V
$U_i \leq 60$	1 000	1 415
$60 < U_i \leq 300$	1 500	2 120
$300 < U_i \leq 690$	1 890	2 670
$690 < U_i \leq 800$	2 000	2 830
$800 < U_i \leq 1\,000$	2 200	3 110
$1\,000 < U_i \leq 1\,500$ ^a	-	3 820

^a For d.c. only.

^b Test voltages based on 6.1.3.4.1, fifth paragraph, of IEC 60664-1.

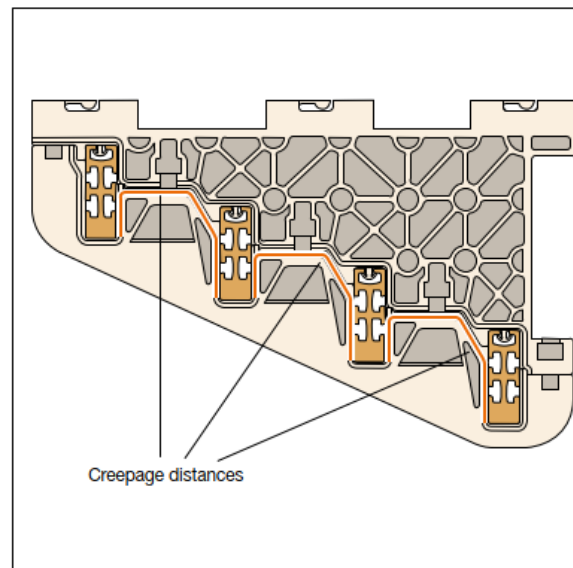


Table 9 – Power-frequency withstand voltage for auxiliary and control circuits (10.9.2)

Rated insulation voltage U_i (line to line)	Dielectric test voltage a.c. r.m.s.
V	V
$U_i \leq 12$	250
$12 < U_i \leq 60$	500
$60 < U_i$	See Table 8

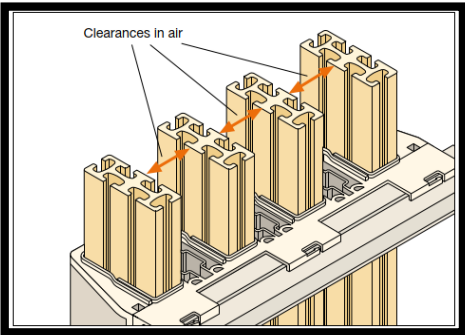
Product certification – Switchgear & controlgear assemblies

8) Dielectric Properties - Clause:10.9

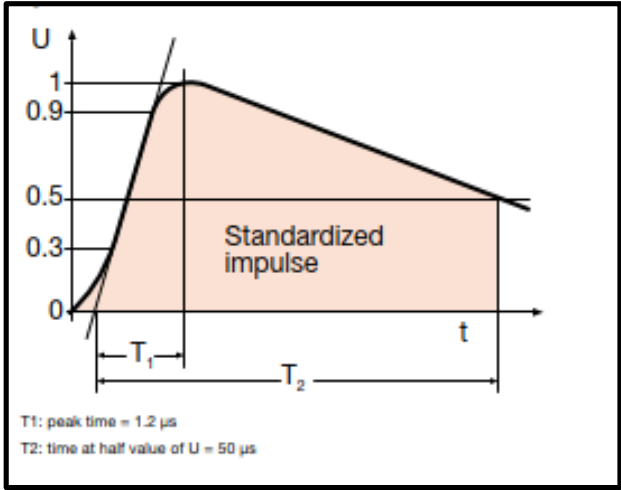
Impulse withstand voltage: Clause:10.9.3

Table 10 – Impulse withstand test voltages (10.9.3)

Rated impulse withstand voltage U_{imp} kV	Test voltages and corresponding altitudes during test									
	$U_{1,2/50}$, a.c. peak and d.c. kV					a.c. r.m.s. kV				
	Sea level	200 m	500 m	1 000 m	2 000 m	Sea level	200 m	500 m	1 000 m	2 000 m
2,5	2,95	2,8	2,8	2,7	2,5	2,1	2,0	2,0	1,9	1,8
4,0	4,8	4,8	4,7	4,4	4,0	3,4	3,4	3,3	3,1	2,8
6,0	7,3	7,2	7,0	6,7	6,0	5,1	5,1	5,0	4,7	4,2
8,0	9,8	9,6	9,3	9,0	8,0	6,9	6,8	6,6	6,4	5,7
12,0	14,8	14,5	14,0	13,3	12,0	10,5	10,3	9,9	9,4	8,5



Test:
Requires the application of the impulse withstand voltage 1,2/50µs.



Design rule:
Shall confirm that the clearances between all the live parts and the parts subject to the risk of discharge are at least 1.5 times the values specified in table below

Rated impulse withstand voltage U_{imp} kV	Minimum clearance in air mm
≤ 2,5	1,5
4,0	3,0
6,0	5,5
8,0	8,0
12,0	14,0

* Based on inhomogeneous field conditions and pollution degree 3.

Product certification – Switchgear & controlgear assemblies

9) Verification of temperature rise: Clause:10.10

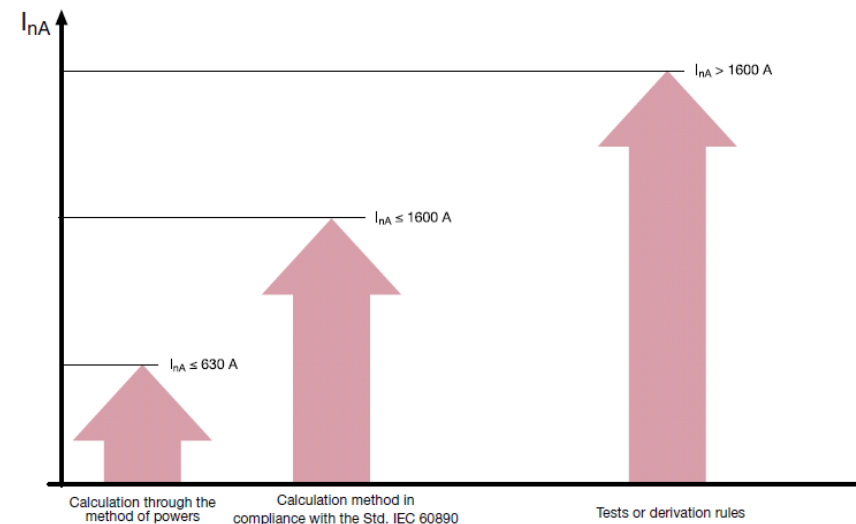
Design Verification – Temperature rise

No.	Characteristic to be verified	Clauses or subclauses	Verification options available		
			Testing	Comparison with a reference design	Assessment
10	Temperature-rise limits	10.10	YES	YES	YES

Verification by calculation:

Method of power (up to 630A)

Calculation according to IEC TR 60890 (up to 1600A)



Product certification – Switchgear & controlgear assemblies

9) Verification of temperature rise - Clause:10.10

Table 6 – Temperature-rise limits (9.2)

Parts of ASSEMBLIES	Temperature rise K
Built-in components ^a	In accordance with the relevant product standard requirements for the individual components or, in accordance with the component manufacturer's instructions ^f , taking into consideration the temperature in the ASSEMBLY
Terminals for external insulated conductors	70 ^b
Busbars and conductors	Limited by ^f : – mechanical strength of conducting material ^g; – possible effect on adjacent equipment; – permissible temperature limit of the insulating materials in contact with the conductor; – effect of the temperature of the conductor on the apparatus connected to it; – for plug-in contacts, nature and surface treatment of the contact material
Manual operating means: – of metal – of insulating material	15 ^c 25 ^c
Accessible external enclosures and covers: – metal surfaces – insulating surfaces	30 ^d 40 ^d
Discrete arrangements of plug and socket-type connections	Determined by the limit for those components of the related equipment of which they form part ^e
NOTE 1 The 105 K relates to the temperature above which annealing of copper is likely to occur. Other materials may have a different maximum temperature rise. NOTE 2 The temperature rise limits given in this table apply for a mean ambient air temperature up to 35 °C under service conditions (see 7.1). During verification a different ambient air temperature is permissible (see 10.10.2.3.4).	

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Table 7 – Temperature-rise limits for terminals and accessible parts

Description of part [*]	Temperature-rise limits ^{**} K
– Terminals for external connections	80
– Manual operating means: - metallic - non-metallic	25 35

Ambient temperature as per IEC 60947-1 is -5 Deg C +40Deg C

Maximum absolute temperature = rise (K) + Ambient 40 Deg C
120C = 80K+40C

Ambient temperature as per IEC 61439-2 is -10 Deg C +40Deg C

Maximum absolute temperature = rise (K) + Ambient 40 Deg C
120C = 70K+50C

^a The term "built-in components" means:

- conventional switchgear and controlgear;
- electronic sub-assemblies (e.g. rectifier bridge, printed circuit);
- parts of the equipment (e.g. regulator, stabilized power supply unit, operational amplifier).

Product certification – Switchgear & controlgear assemblies

9) Verification of temperature rise - Clause:10.10

Temperature rise limit (ref. para. 10.10.2.3.5 & 10.10.2.3.7 of the standard)

➡ Verification of complete assembly 10.10.2.3.5 :

➡ 6300A, 5500A, 5000A, 4000A, 3200A, 2500A, 1600A, 1250A, 800A & 400A at normal ambient (35 deg C) and 50 Deg C with different IP rating.

➡ Verification of Main busbar 10.10.2.3.7 :

➡ 6700A, 5800A, 5000A, 3200A, 2500A, 1600A, 1250A, 800A & 400A at normal ambient (35 deg C) and 50 Deg C with different IP rating



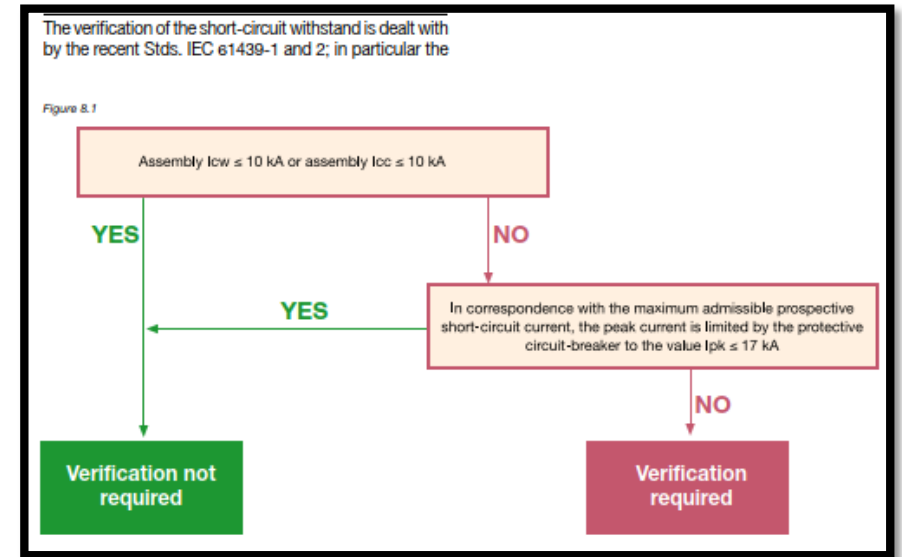
Product certification – Switchgear & controlgear assemblies

10) Short-circuit test: Clause : 10.11

Table 7 – Values for the factor n^a (9.3.3)

r.m.s. value of short-circuit current kA	$\cos \varphi$	n
$I \leq 5$	0,7	1,5
$5 < I \leq 10$	0,5	1,7
$10 < I \leq 20$	0,3	2
$20 < I \leq 50$	0,25	2,1
$50 < I$	0,2	2,2

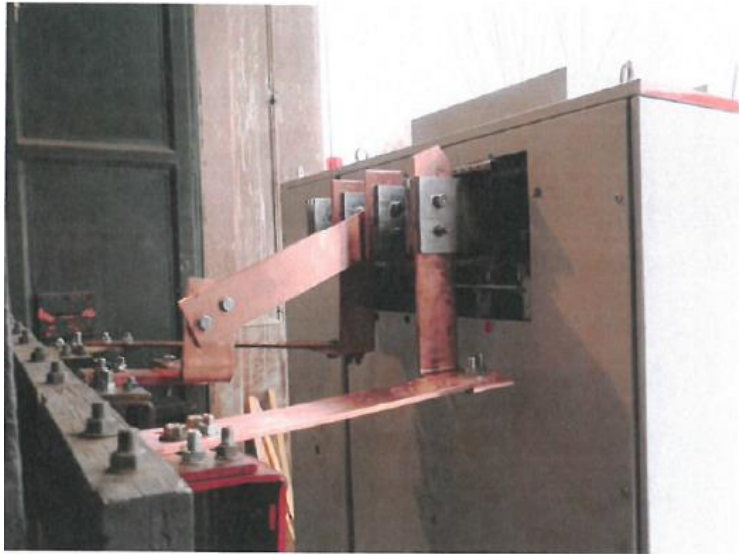
^a Values of this table represent the majority of applications. In special locations, for example in the vicinity of transformers or generators, lower values of power factor may be found, whereby the maximum prospective peak current may become the limiting value instead of the r.m.s. value of the short-circuit current.



Product certification – Switchgear & controlgear assemblies

10) Short-circuit test: Clause : 10.11

- ➔ Inc & O/G circuits (10.11.5.3.2):
 - ➔ Icc 36kA (400A & 800A)
 - ➔ Icc 50kA (1250A, 1600A, 2500A & 3200A)
 - ➔ Icc 65kA (2500A & above)
- ➔ withstand capacity (10.11.5.3.3 & 10.11.5.3.5):
 - ➔ Icw up to 100kA (Inc, MBB & VBB – 3Ph-Ph)
 - ➔ 60% of above for N-Ph



Product certification – Switchgear & controlgear assemblies

11) Electro-magnetic compatibility: Clause:10.12

- No tests are required if the assemblies does not incorporate electronic circuits
- Assemblies incorporating electronic circuits, tests shall be made to the relevant environment A or B
- EMC for the LVB product installed in the ArTu K is verified – Pass.
- All ABB SACE LVB products specially with Electronic releases are tested and comply to EMC requirements.

Environment A: relates to a power network supplied from a high or medium voltage transformer dedicated to the supply of an installation feeding manufacturing or similar plant, and intended to operate in or in proximity to industrial locations.

Environment B: relates to low-voltage public mains networks or apparatus connected to a dedicated DC source which is intended to interface between the apparatus and the low voltage public mains network.

12) Mechanical operation: Clause:10.13

- For parts, which need verification by test (see 8.1.5), satisfactory mechanical operation shall be verified after installation in the ASSEMBLY. The number of operating cycles shall be 200.

ArTu K

Perfect integration of ABB solutions



Perfect fitting of ABB solutions for commercial applications up to 6700A**

- IEC 61439 1-2 type tested
- European design and quality
- Adapted to southern Gulf requirements, ADQCC certified*
- Fully compatible with ABB most advanced devices and communication solutions

ABB solutions: the convenient and safe choice for your projects!



Fixed motor control solutions



Air circuit breakers Emax 2



Molded case circuit breakers Tmax T, XT



Din rail products and front panel instruments



ABB Ability solutions



* Obtained up to 5500A

** MBB rating

ArTu K

Standards and installation



- Obtained up to 5500A

- IEC 61439-1/2 100% certified by DEKRA @50°C ambient temperature
- ADQCC * and AADC certifications
- Arc fault and Seismic testing as per global Standards
- Painted internal parts
- Protection degrees IP43 and IP54. Front protection IP65.
- Blind doors and glass doors available
- Maximum robustness up to IK10
- Rear access as standard / front on option
- For indoors installation
- Pollution degree 3

ArTu K

Electrical features



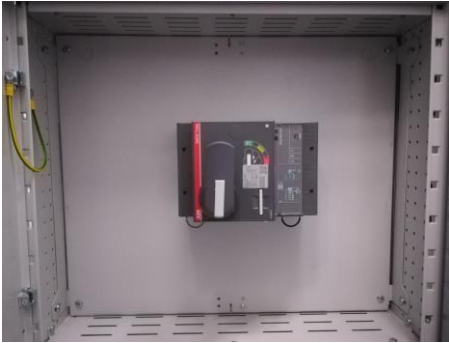
- Suitable for 690V AC

- Suitable for 3P/4P networks
- Frequency 50Hz
- Ratings up to 6700A IP43, 5000A IP54
- Tested at 415 VAC *
- Short circuit up to I_{cw} 100 kA/1s, 65kA/3s & 50kA/3s
- Impulse voltage up to 12 kV
- Insulation voltage up to 1000V
- Segregation up to form 4b type 7 (IEC + BS)

ArTu K

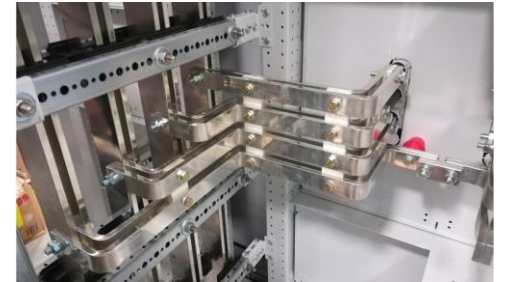
Focus on...

Kits



Easiness

- Strong fitting of any device, from Din rail to incomers up to 6300A
- Pre-punched kit available for each device
- Modularity of the functional frame
- No live parts visible at front side of the compartment (no local shroudings required)



Neat design

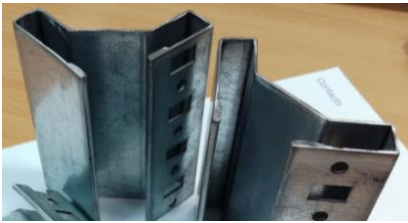
- Devices connected with rear access for a clear switchboard
- Easy wiring
- Form of segregation well maintained



ArTu K

Focus on...

Structure



Frame

- Robust IK10 frame with 8 folds design for maximum robustness
- The frame is proved to be more robust than common market solutions by a 3rd party study.



Claddings

- Glass door for optimal visibility of the devices status while maintaining full protection against unwanted access.
- IP65 on front, protecting against water jets



Ventilations

- Back ventilations to increase performance at high ambient temperature
- IP43 and IP54 versions



Foam gasketing

- Foam gasketing on claddings to reach protection degrees IP43/54/65.
- Long lifetime, no peeling
- High quality contact

ArTu K

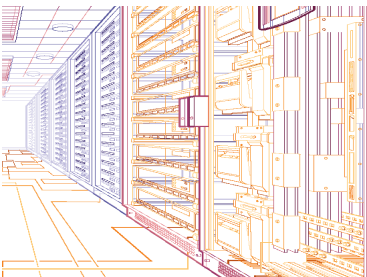
Focus on...

Busbar and segregations



Certification

- Numerous system certifications from 400A to 6700A @50°C to match your needs with less copper
- Certified with busducts



Above Standard segregations

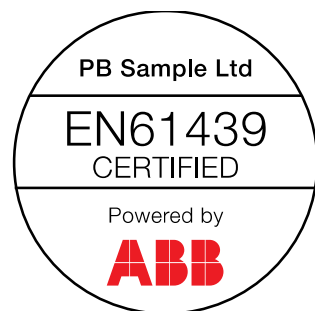
- Form 2, 3b, 4b available as per IEC 61439 1-2
- Form 4b type 7 available as per BS EN 61439 for increased safety

ArTu K

Offering through Authorized Channel Partners

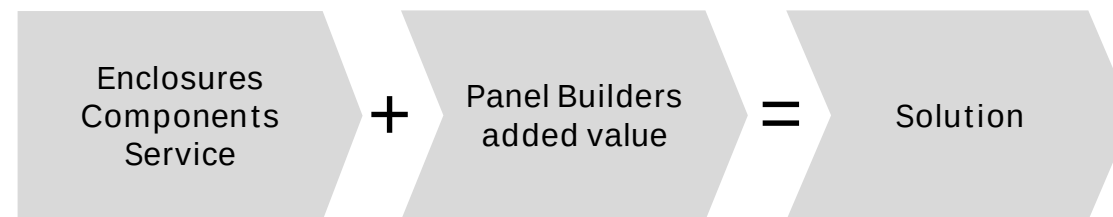


Up to 4000A
or 6700A



A strong grid of Channel Partners to deliver advanced solutions

- Strong experience on projects realization with ABB offer
- Deep knowledge on IEC 61439 1-2 Standard and processes
- Fully trained to implement ABB offer
- Continuous improvements and trainings



The right ABB components + Panel Builder's added value = best performance

ArTu K

Additional certification

Arc fault certification

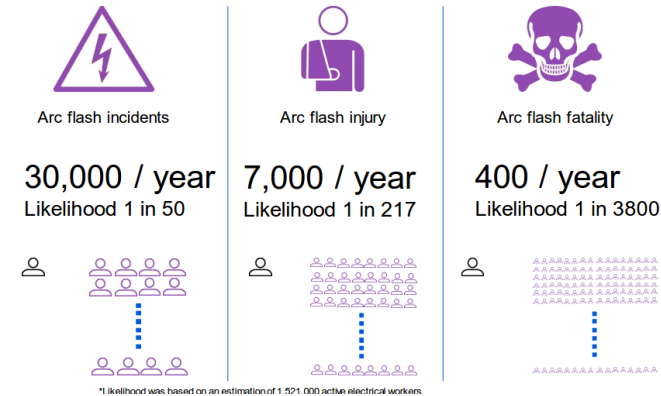
- ArTu K provides the maximum level of safety with Internal Arc Test certification following the highest criteria defined by the latest IEC TR 61641 International Standard.
- Tested as per the latest version of the standard: edition 3.
- Criteria 1...7, Class C
- Voltage 415V - 65kA/300ms

GULF NEWS EMERGENCIES

Gulf News – 2012
Arc Flash Accident Leaves 4 Dubai Men Injured
Extreme temperature flare caused burns in Dubai airport accident

Gulf News – 2013
Arc Flash at Mall Leaves 3 Workers in Critical Condition
Three workers are fighting for their lives at Rashid Hospital after suffering severe electrical burns in a site accident behind Wafi Mall

USA statistics:

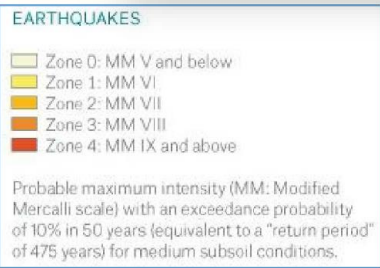


Additional certification

Seismic certification

- ArTu K provides an advanced protection in case of seismic events with performances that are certified by the most stringent International Standards.
- Cert. 0.5g, Richter 7, Mercalli IX, Zone 4 by Virlab / Cisa in Europe

Seismic Zone IEC 68-3-3	Scale of Richter Magnitude	Scale of Mercalli Magnitude	Phenomena	Maximum acceleration (g)	Free energy (J)
AG5	9	XII	Exceptionally catastrophic	1.53	$>10^{17}$
	8	XI	Catastrophic	1.02	
	7	X	Very destructive	0.51-1.02	
AG3	6	IX	Destructive	0.2-0.5	$10^{14} - 5 \times 10^{15}$
	5	VIII	Damaging	0.12-0.2	
	4	VII	Very strong		
AG2	3	VI	Strong		$10^{11} - 5 \times 10^{12}$
	2	V	Fairly strong		
	1	IV	Not strong		
	0	III	Weak		$10^8 - 5 \times 10^9$
		II	Very weak		
	1	I	Registered by instrument		$5 \times 10^6 - 10^8$



Qatar, Bahrain, part of UAE and Oman are classified as Zone 1.
Northern UAE and northern Oman are classified as Zone 2.

- ABB certification (standard configuration)
- Maximum risk in the area as per Munich Re

ArTu K

Take away message!

- Maximum performance on the market
- Fully meet the new regulation and requirements
- Maximum short circuit rating of 100kA/1s
- One platform from 400A to 6700A in three performance levels
- Increased Icw rating for MBB 65kA/3s
- Products with third party type test certificate without any “certificate of conformity”
- Certified to IP54 as per IEC60529
- External door (double door) will ensure 100% IP degree of protection, if any type of component installed on internal door.

ArTu K

There is always someone do it cheaper with specification



and ABB offers meets the requirement in sincere test method



TUESDAY, JUNE 23, 2020

IEC 60947-2 Standard

Standard for Low Voltage Circuit Breakers

Product Marketing – Electrification business – Smart Power, United Arab Emirates



Standard Guidelines

IEC60947-2

- Circuit Breaker Standard, for industrial application
- Definitions for MCCBs and ACBs
- Choice criteria based on rated and limit values

Standard for LV apparatus

IEC 60947 Standard for industrial application

IEC 60947

- IEC 60947-1 Part 1: General rules
- IEC 60947-2 Part 2: Circuit breakers
- IEC 60947-3 Part 3: Switch disconnectors
- IEC 60947-4-1 Part 4: Contactors
- IEC 60947-5-1 Part 5: Control circuit devices
- IEC 60947-6-1 Part 6: Multifunction devices
- IEC 60947-7-1 Part 7: Auxiliary materials
- IEC 60947-8 Part 8: Control units for built-in thermal protection (PTC)

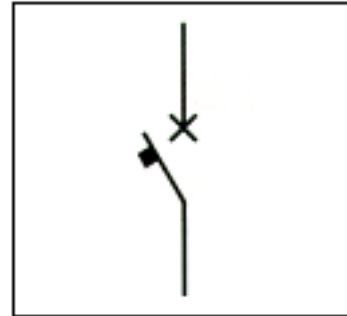
IEC Standard definitions

IEC 60947-2

Circuit Breaker

A mechanical switching device capable of breaking, carrying and making currents under normal circuit conditions and also making, carrying, for a specified time, and breaking currents under specified abnormal circuit conditions such as those of short-circuit.

- | | |
|-------------|----------------------|
| ▪ BREAKING | Breaking Capacity |
| ▪ WITHSTAND | Short time withstand |
| ▪ MAKING | Making Capacity |



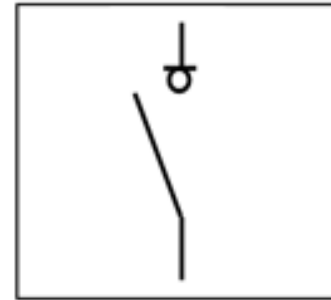
IEC Standard definitions

IEC 60947-3

Switch Disconnecter

A mechanical switching device capable of breaking, making and carrying currents under normal circuit conditions but only making and carrying, for a specified time, currents under specified abnormal circuit conditions such as those of short-circuit.

- ~~BREAKING~~ **NO** ~~Breaking Capacity~~
- WITHSTAND Short time withstand
- MAKING Making Capacity



IEC Standard definitions

IEC 60947-2

Moulded Case Circuit Breaker (MCCB)

A circuit breaker having a supporting housing of moulding insulating material, forming an integral part of the circuit breaker (Tmax-XT & Formula).



IEC Standard definitions

IEC 60947-2

Air Circuit Breaker (ACB)

A circuit breaker having a supporting housing of moulding insulating material and a metallic frame, forming an integral part of the circuit breaker (Emax 2).

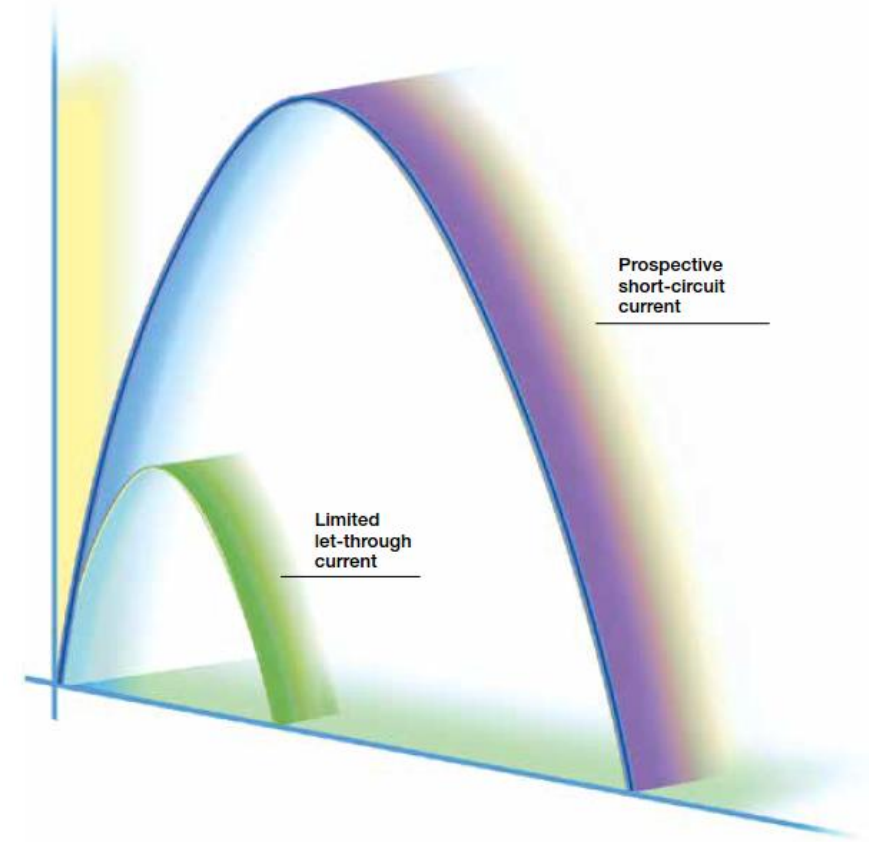


IEC Standard definitions

IEC 60947-2

Current Limiting Circuit Breaker

A circuit breaker that, within a specified range of current, prevents the let-through current reaching the prospective peak value and which limits the let-through energy (I^2t) to a value less than the let-through energy of a half-cycle wave of the symmetrical prospective current.



IEC Standard definitions

Choice criteria

- Rated values (I_u, U_e) 
- Limit values ($I_{cu}, I_{cs}, I_{cw}, I_{cm}$)
- Insulation values (U_i, U_{imp})

IEC Standard definitions

IEC 60947-1

Rated uninterrupted current I_u

The rated uninterrupted current of an equipment is a value of current, stated by the manufacturer, that the equipment can carry in uninterrupted duty (at 40 °C)

The rated uninterrupted current I_u is different from the rated current I_n , which is the rated current of the thermomagnetic or electronic trip unit and is lower or equal to I_u .

Some factors may reduce the I_u of a circuit breaker like breaker type, temperature, altitude or frequency.

Rating plugs

Type of circuit-breaker	Rated current I_n	I_n [A]	400	630	800	1000	1250	1600	2000	2500	3200	4000	5000	6300
E1B	800													
	1250													
	1600													
E1N	800													
	1250													
	1600													
E2B	1600													
	2000													
E2N	1250													
	1600													
	2000													
E2S	800													
	1250													
	1600													
	2000													
E2L	1250													
	1600													
E3N	2500													
	3200													
E3S	1250													
	1600													
	2000													
	2500													
	3200													

IEC Standard definitions

IEC 60947-1

Rated operational voltage U_e

The rated operational voltage of an equipment is a value of voltage which, combined with a rated operational current, determines the application of the equipment and to which the relevant tests and the utilization categories are referred.

IEC Standard definitions

IEC 60947-1

Rated operational voltage U_e

Breaking capacity is always referred to the operational voltage; the breaking capacity decreases when the voltage increases.

		XT1					
Size ^(G2.1)		[A]	160				
Poles		[No.]	3, 4				
Rated service voltage, U_e ^(G2.4)	(AC) 50-60Hz	[V]	690				
	(DC)	[V]	500				
Rated insulation voltage, U_i ^(G2.5)		[V]	800				
Rated impulse withstand voltage, U_{imp} ^(G2.6)		[kV]	8				
Versions			Fixed, Plug-in ⁽²⁾				
Breaking capacities according to IEC 60947-2			B	C	N	S	H
Rated ultimate short-circuit breaking capacity, I_{cu} ^(G2.7)							
Icu @ 220-230-240V 50-60Hz (AC)		[kA]	25	40	65	85	100
Icu @ 380V 50-60Hz (AC)		[kA]	18	25	36	50	70
Icu @ 415V 50-60Hz (AC)		[kA]	18	25	36	50	70
Icu @ 440V 50-60Hz (AC)		[kA]	15	25	36	50	65
Icu @ 500V 50-60Hz (AC)		[kA]	8	18	30	36	50
Icu @ 525V 50-60Hz (AC)		[kA]	6	8	22	35	35
Icu @ 690V 50-60Hz (AC)		[kA]	3	4	6	8	10

IEC Standard definitions

IEC 60947-1

Rated operational voltage U_e

Some factors may reduce the U_e of a circuit breaker

No nominal performance derating for use up to an altitude of 2000m. Above 2000m, the properties of the atmosphere (composition of the air, dielectric strength, cooling power and pressure) change, having an impact on the main parameters which define the circuit-breaker. The table below gives the changes to the main performance parameters.

Altitude		2000m	3000m	4000m	5000m
Rated employ voltage, U_e	[V]	690	600	540	470
Rated uninterrupted current	%	100	98	93	90

IEC Standard definitions

Choice criteria

- Rated values (I_u , U_e)
- **Limit values (I_{cu} , I_{cs} , I_{cw} , I_{cm})** 
- Insulation values (U_i , U_{imp})

IEC Standard definitions

IEC 60947-2

Rated ultimate short circuit breaking capacity I_{cu}

Breaking capacity according to a specified test sequence, do not include after the short circuit test, the capability of the circuit breaker to carry its rated current continuously.

- test sequence: O - 3 min - CO
- dielectric withstand at $2 \times U_e$
- verification of overload release at $2.5 \times I_1$

T5N400 PR221DS-LS/I In 320

I_{cu}



IEC Standard definitions

IEC 60947-2

Rated service short circuit breaking capacity I_{cs}

Breaking capacity according to a specified test sequence. Include after the short circuit test, the capability of the circuit breaker to carry its rated current continuously

- test sequence: O - 3 min - CO - 3 min – CO (25%-50%-75%-100% of I_{cu})
- operational performance capability: (5% of N° operating cycle for operational performance capability- table 8 of IEC60947-2)
- dielectric withstand at $2 \times U_e$
- verification of temperature rise at I_u (the temperature doesn't exceed the limits of table 7 of IEC60947-2)
- verification of overload release at $1.45 \times I_1$

IEC Standard definitions

IEC 60947-2

Relation between I_{cs} and I_{cu}

The service breaking capacity I_{cs} can be expressed as a value of breaking current, in kA;
A percentage of I_{cu} , rounded up to the lowest whole number, in accordance with the table (for example $I_{cs} = 25\% I_{cu}$).

% of I_{cu}
25
50
75
100

Standard ratios between I_{cs} and I_{cu}

IEC Standard definitions

IEC 60947-2

When is Icu required?

- Where continuity of service is not a fundamental requirement.
- For protection of single terminal load.
- For motor protection.
- Where maintenance work is easily carried out without much disruption.

Generally for circuit breaker installed on terminals part of plant.

When is Ics required?

- Where continuity of service is a fundamental requirement.
- For installation in power center.
- Where is more difficult to make maintenance.
- When is difficult to manage spare breakers.

Generally for installation in main distribution board immediately downstream transformer or generator.

IEC Standard definitions

IEC 60947-2

Rated short-time withstand current I_{cw}

The rated short-time withstand current of a circuit-breaker is the value of short-time withstand current assigned to that circuit-breaker by the manufacturer under the specified test conditions.

For a.c., the value of this current is the r.m.s. value of the a.c. component of the prospective short-circuit current, assumed constant during the short-time delay.

The short-time delay associated with the rated short-time withstand current shall be at least 0.05s. Preferred values being as follows:

0.05 s - 0.1 s – 0.25 s – 0.5 s – 1 s

IEC Standard definitions

IEC 60947-2

Rated short-time withstand current I_{cw}

Minimum values if rated short-time withstand current I_{cw} (Table 3)

Rated current I_n	Rated short-time withstand current I_{cw} – Minimum values
A	kA
$I_n \leq 2\,500$	12 I_n or 5 kA, whichever is the greater
$I_n > 2\,500$	30 kA

IEC Standard definitions

IEC 60947-2

Rated short-time withstand current I_{cw}

Selectivity categories

- Selectivity category B comprises circuit-breakers providing selectivity by having a short-time withstand current rating and an associated short-time delay.
Selectivity of circuit-breakers of selectivity category B is not necessarily ensured up to the ultimate short-circuit breaking capacity but up to the value specified in Table 3.
- Selectivity category A comprises of all other circuit-breakers.
These circuit-breakers may provide selectivity under short-circuit conditions by other means.

IEC Standard definitions

IEC 60947-2

Rated short-circuit making capacity I_{cm}

Making capacity for which the prescribed conditions according to a specified test sequence include the capability of the circuit breaker to make the peak current corresponding to that rated capacity at the appropriate applied voltage.

It is always necessary to verify that:

$$I_{cm} \geq I_{peak}$$



IEC Standard definitions

IEC 60947-2

Rated short-circuit making capacity I_{cm}

It is the maximum prospective peak current which the circuit-breaker must be able to make

$$I_{cm} = n \times I_{cu}$$

Short-circuit breaking capacity kA r.m.s.	Power factor	Minimum value required for n $n = \frac{\text{short - circuit making capacity}}{\text{short - circuit breaking capacity}}$
$I \leq 1,5$	0,95	1,41
$1,5 < I \leq 3$	0,9	1,42
$3 < I \leq 4,5$	0,8	1,47
$4,5 < I \leq 6$	0,7	1,53
$6 < I \leq 10$	0,5	1,7
$10 < I \leq 20$	0,3	2,0
$20 < I \leq 50$	0,25	2,1
$50 < I$	0,2	2,2

T5N400 PR221DS-LS / I In 320

$$I_{cu} \text{ 36kA} \rightarrow I_{cm} \text{ 75.6 kA @415V}$$

IEC Standard definitions

Choice criteria

- Rated values (I_u , U_e)
- Limit values (I_{cu} , I_{cs} , I_{cw} , I_{cm})
- **Insulation values (U_i , U_{imp})** 

IEC Standard definitions

IEC 60947-1

Rated insulation voltage U_i

The rated insulation voltage of an equipment is the value of voltage to which dielectric tests and creepage distances are referred.

It shall be always verified that:

$$U_e \leq U_i$$

IEC Standard definitions

IEC 60947-1

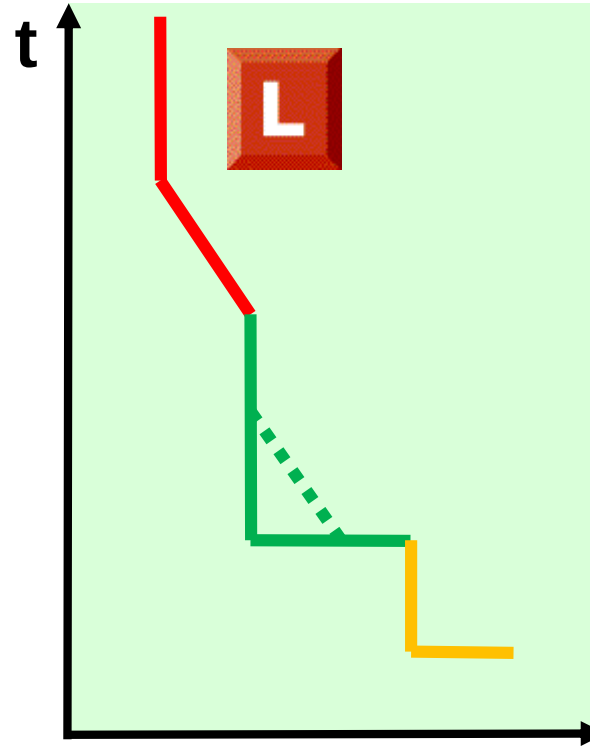
Rated Impulse withstand voltage U_{imp}

Rated impulse withstand voltage, referred to as U_{imp} , is the peak value of an impulse voltage of prescribed form and polarity that the equipment is capable of withstanding without failure under specified test conditions, and to which the value of the clearances are referred.

U_{imp} is equal to:
8kV for MCCBs
12kV for ACBs

It is necessary to verify that
 $U_{imp} \geq$ transient overvoltage in plant.

IEC 60947-2

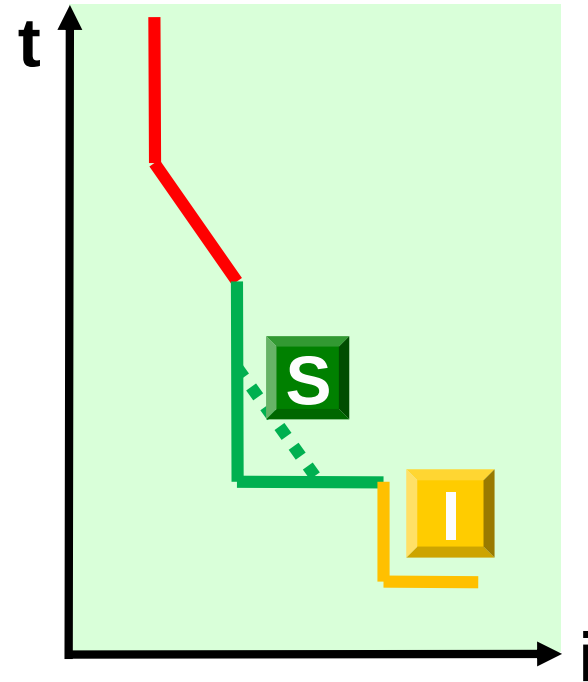
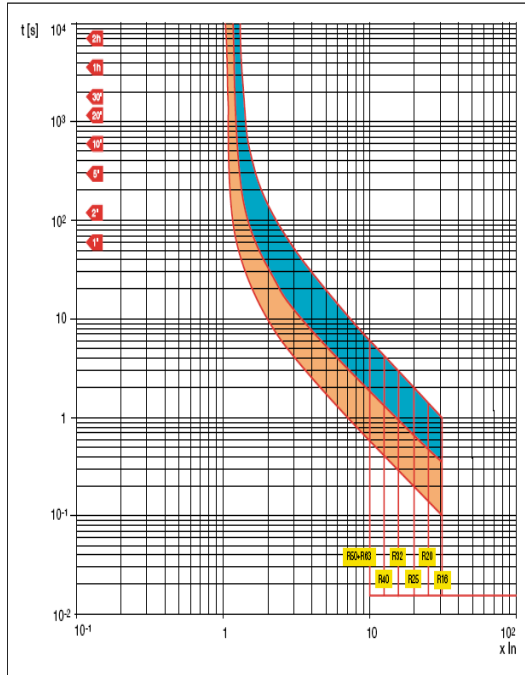


IEC Standard definitions

IEC 60947-2

Short circuit protection

The operation of short-circuit releases shall be verified at 80% and 120% of the short-circuit setting of the release. The test current shall have no asymmetry.



IEC Standard definitions

IEC 60947-2

Type Test

The tests to verify the characteristics of circuit breakers are:

- type tests carried out on samples:

Test
Temperature-rise
Tripping limits and characteristics
Dielectric properties
Operational performance capability
Overload performance (where applicable)
Short-circuit breaking capacities
Short-time withstand current (where applicable)
Performance of integrally fused circuit-breakers

IEC Standard definitions

IEC 60947-2

Temperature-rise limits and accessible parts

Description of part ^a	Temperature-rise limits ^b K
– Terminals for external connections	80
– Manual operating means: metallic	25
non-metallic	35
– Parts intended to be touched but not hand-held: metallic	40
non-metallic	50
– Parts which need not be touched for normal operation: metallic	50
non-metallic	60
^a No value is specified for parts other than those listed but no damage should be caused to adjacent parts of insulating materials.	
^b The temperature-rise limits specified are not intended to apply to a new sample, but are those applicable to the temperature-rise verifications during the appropriate test sequences specified in Clause 8.	

IEC Standard definitions

IEC 60947-2

Type Test



IEC Standard definitions

CE Marking

- According to european directives:
 - Low Voltage Directive 2014/35/EC
 - Electromagnetic Compatibility 2014/30/EC





Low voltage selectivity

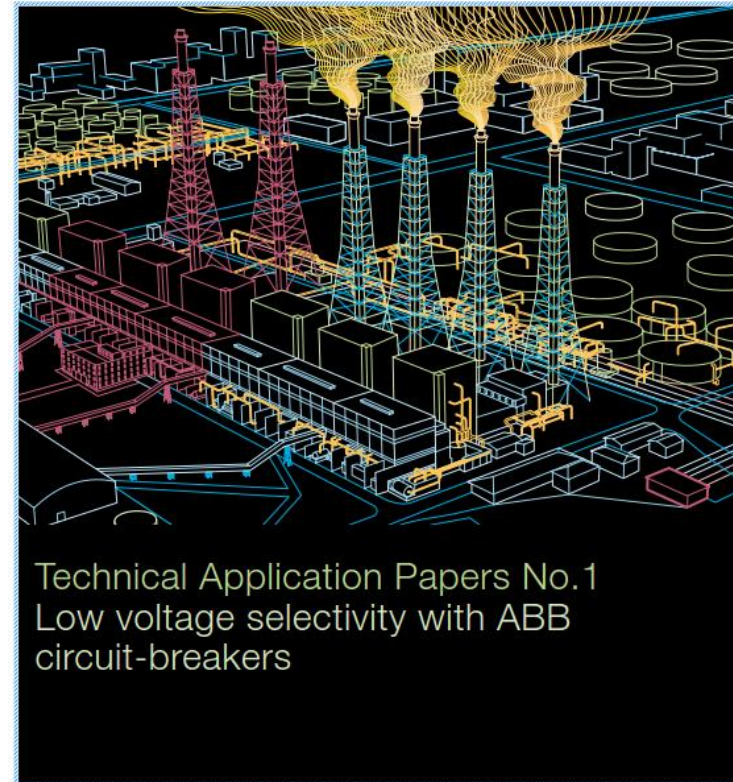
Selectivity definitions & standards

Low voltage selectivity

Selectivity definitions & standards

Agenda

- Definitions & Standards
- Selectivity Techniques
- Back-Up Protection



Low voltage selectivity

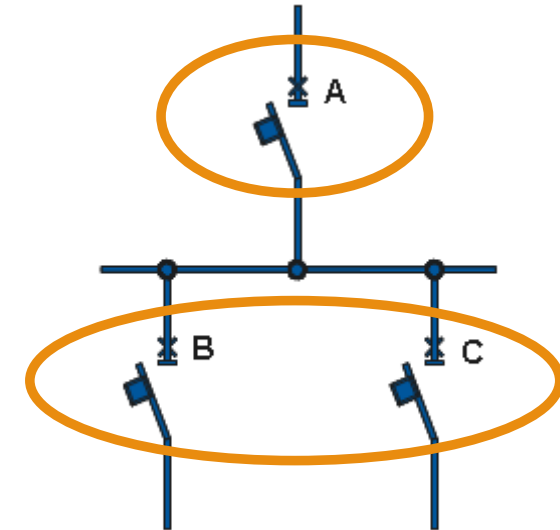
Selectivity definitions & standards

What is selectivity?

Selectivity (or discrimination)

- is a type of coordination of two or more protective devices in series.
- Selectivity is done between one circuit breaker on the supply side and one circuit breaker, or more than one, on the load side.

A is the supply side circuit breaker (or upstream)



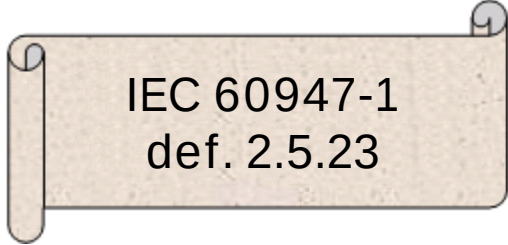
B and C are the load side circuit breakers (or downstream)

Low voltage selectivity

Selectivity definitions & standards

Definition of selectivity

IEC 60947-1 Standard: “Low voltage equipment
Part 1: General rules for low voltage equipment”



IEC 60947-1
def. 2.5.23

“Trip selectivity (for overcurrent) is a coordination between the operating characteristics of two or more overcurrent protection devices, so that, when an overcurrent within established limits occurs, the device destined to operate within those limits trips whereas the others do not trip”

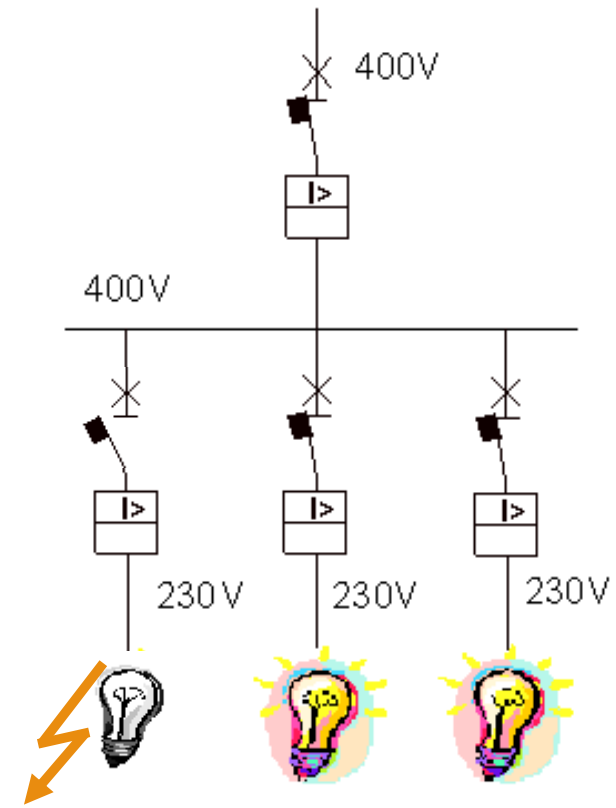
Low voltage selectivity

Selectivity definitions & standards

Example

If selectivity is provided

- When a fault occurs (Overload or Short-circuit)
- The protection device closest to the fault opens.



Low voltage selectivity

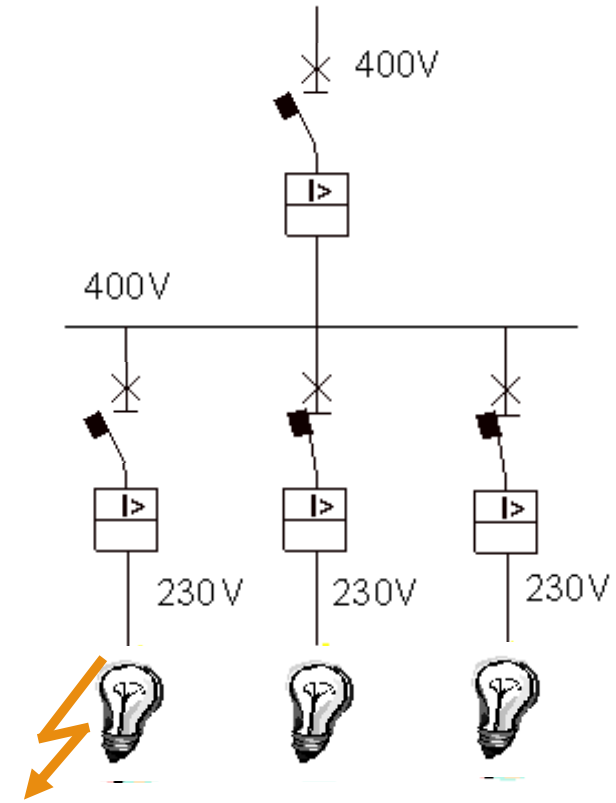
Selectivity definitions & standards

Example

If selectivity is NOT provided

- When a fault occurs (Overload or Short-circuit)
- Both the upstream & downstream protection devices could open.

The whole system will be out of service



Low voltage selectivity

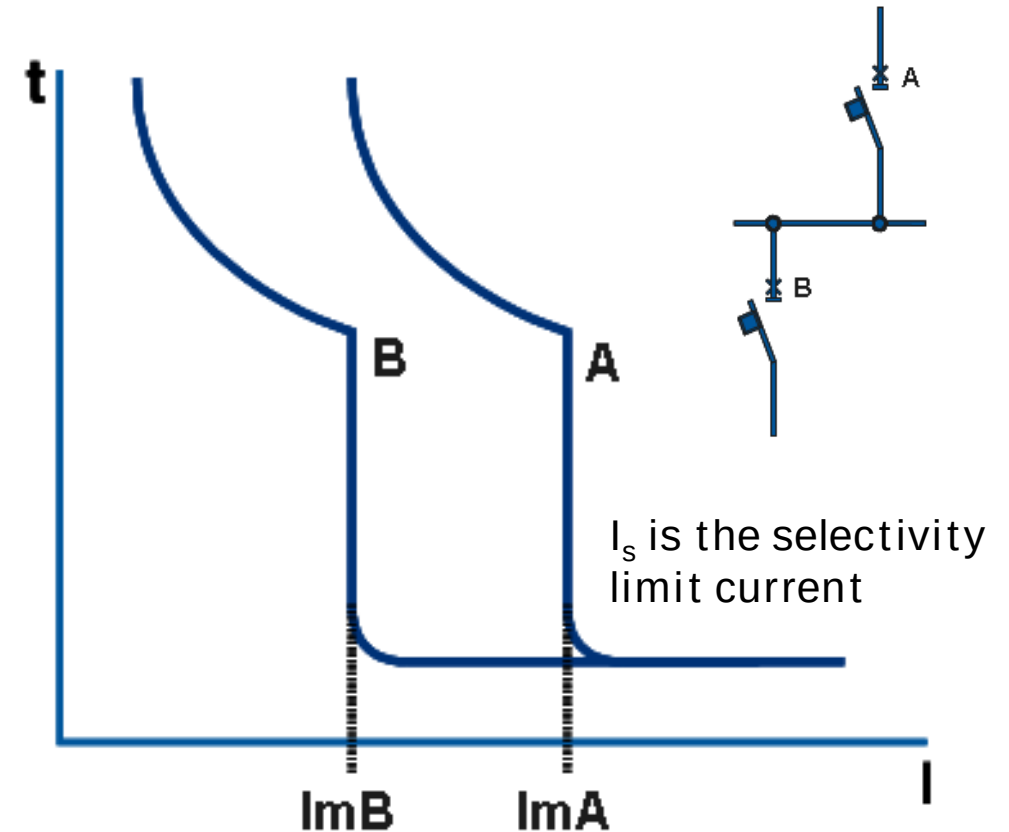
Selectivity definitions & standards

Partial selectivity

“Partial selectivity is an overcurrent selectivity where, in the presence of two protection devices in series, the load side protection device carries out the protection up to a given level of overcurrent, without making the other device trip.”

- B opens only according to fault current lower than a certain current value;
- values equal or greater than I_s will give the trip of both A and B.

$$I_s = I_{mA}$$



Low voltage selectivity

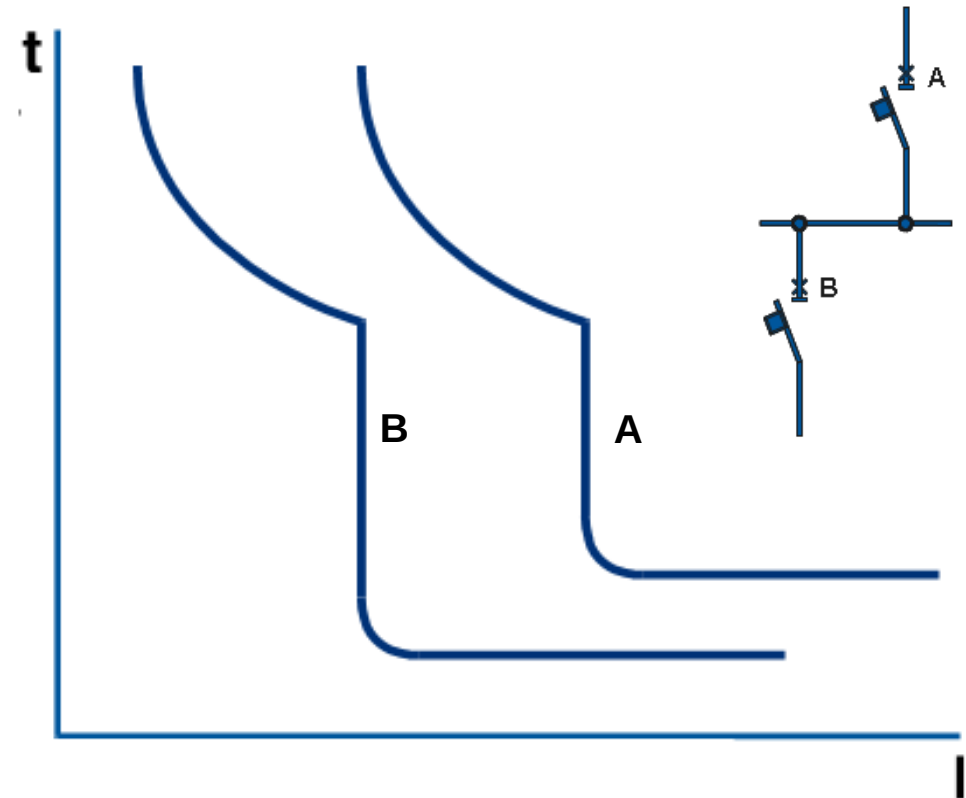
Selectivity definitions & standards

Total selectivity

“Total selectivity is an overcurrent selectivity where, in the presence of two protection devices against overcurrent in series, the load side protection device carries out the protection without making the other device trip.”

- Only B trips for every current value lower or equal to the maximum short-circuit current..

$$I_s = I_k$$



Low voltage selectivity

Selectivity definitions & standards

Partial & Total selectivity

Example

Upstream circuit breaker A :
T4N 250 PR221DS In = 250 (I_{cu} = 36kA)

Downstream circuit breaker B
S 294 C100 (I_{cu} = 15kA)

6kA)

			Supply s.	T4					
			Version	N, S, H, L, V					
			Release	TM, M			EL		
			I _u [A]	250			250		320
Load s.	Char.	I _{cu} [kA]	I _n [A]	160	200	250	160	250	320
S290	C-K	15	80	5	11	T	T	T	T
			100	5*	8	T	12	T	T
	C		125		8*	12		T	T
	D		80	5	11	T	T	T	T
			100		8	T	12	T	T

Low voltage selectivity

Selectivity definitions & standards

Partial & Total selectivity

Example

Upstream circuit breaker A :
T4N 250 PR221DS In = 160 (I_{cu} = 36kA)

Downstream circuit breaker B
S 294 C100 (I_{cu} = 15kA)

6kA)

			Supply s.	T4					
			Version	N, S, H, L, V					
			Release	TM, M			EL		
			I _u [A]	250			↓ 250	320	
Load s.	Char.	I _{cu} [kA]	I _n [A]	160	200	250	160	250	320
S290	C-K	15 →	80	5	11	T	T	T	T
			100	5*	8	T	12	T	T
	C		125		8*	12		T	T
	D		80	5	11	T	T	T	T
			100		8	T	12	T	T

Low voltage selectivity

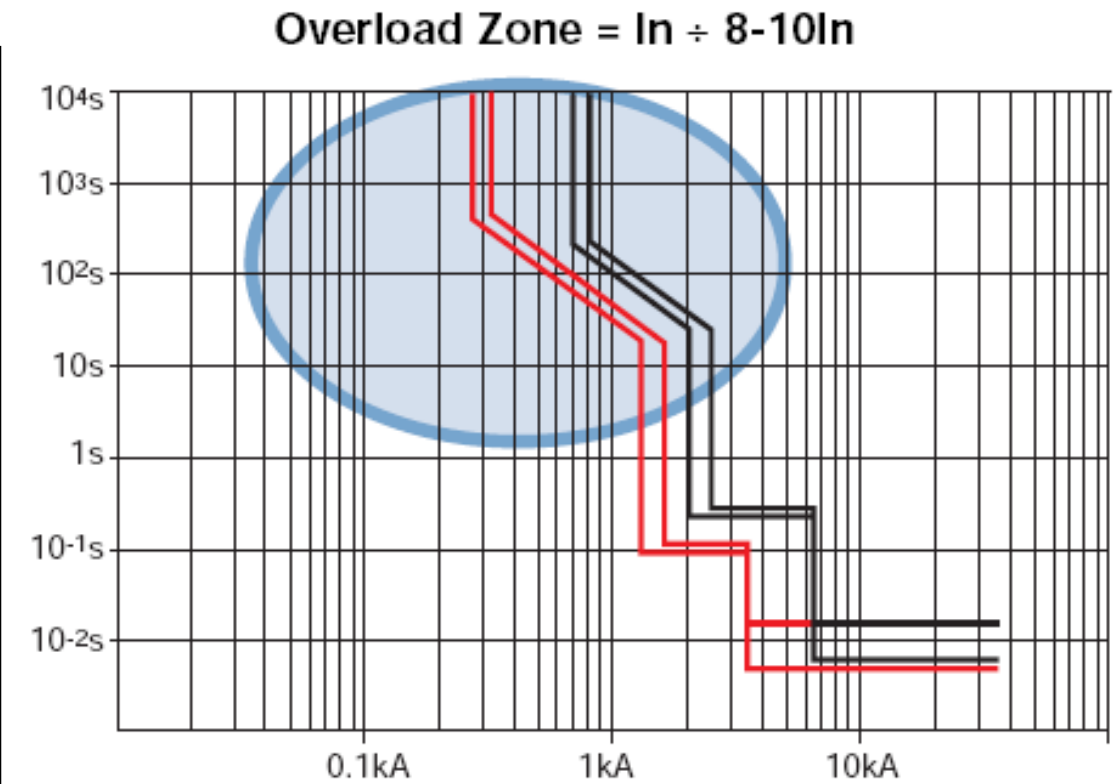
Selectivity definitions & standards

Time-Current Curves

Overload Zone

- Thermal Protection
- L Protection

Time-Current selectivity



Low voltage selectivity

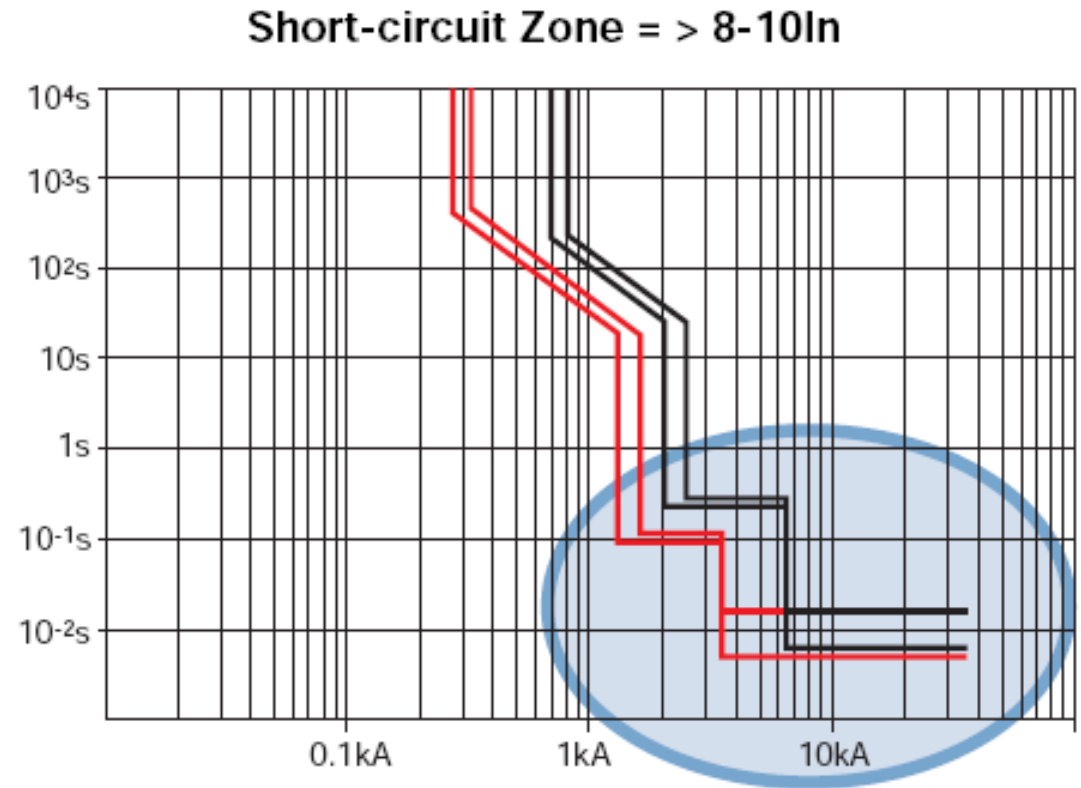
Selectivity definitions & standards

Time-Current Curves

Short-Circuit Zone

- Magnetic protection
- S, D, I & EF Protection

Current, Time, Energy, Zone, Directional & Zone
Directional Selectivity

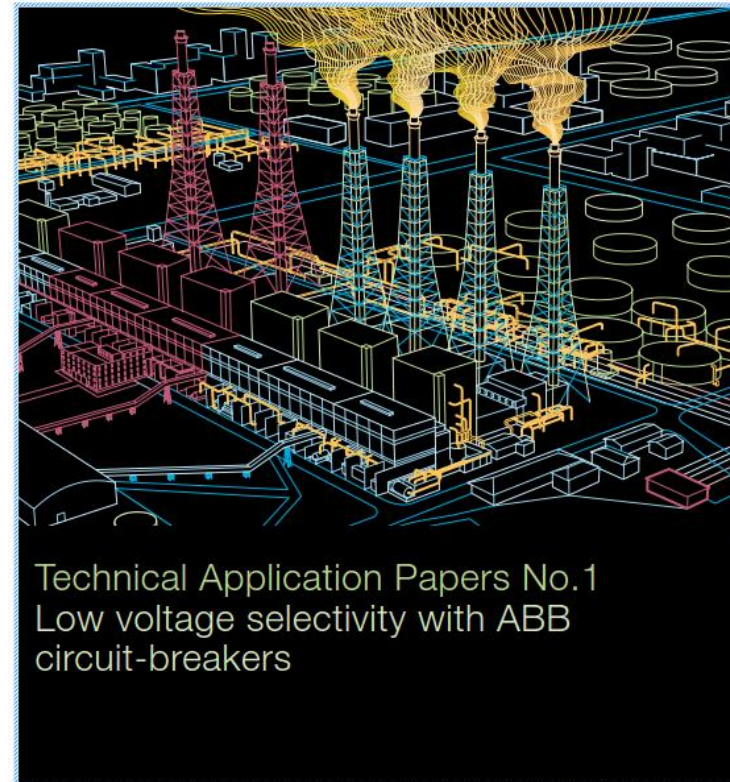


Low voltage selectivity

Selectivity definitions & standards

Agenda

- Definitions & Standards
- Selectivity Techniques
- Back-Up Protection



Low voltage selectivity

Selectivity techniques

Common selectivity techniques

- Current selectivity
- Time selectivity
- Energy selectivity
- Zone (logical) selectivity



Low voltage selectivity

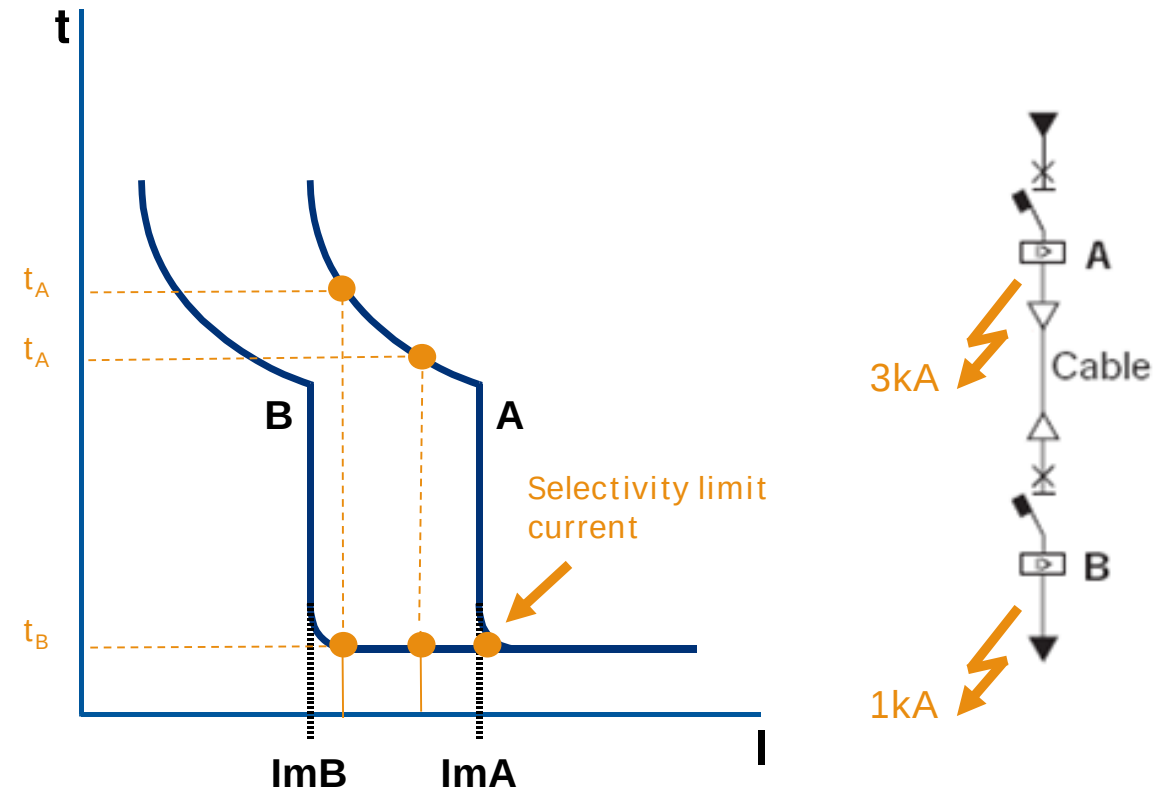
Selectivity techniques

Current selectivity

Basic concept

When the point of fault is closer to the source, the fault current will be higher.

- In order to guarantee selectivity, the protections must be set to different values of current thresholds.
- The ultimate selectivity value is equal to the instantaneous trip threshold of the upstream protection device.
- Other methods are needed to have a total selectivity



Low voltage selectivity

Selectivity techniques

Current selectivity

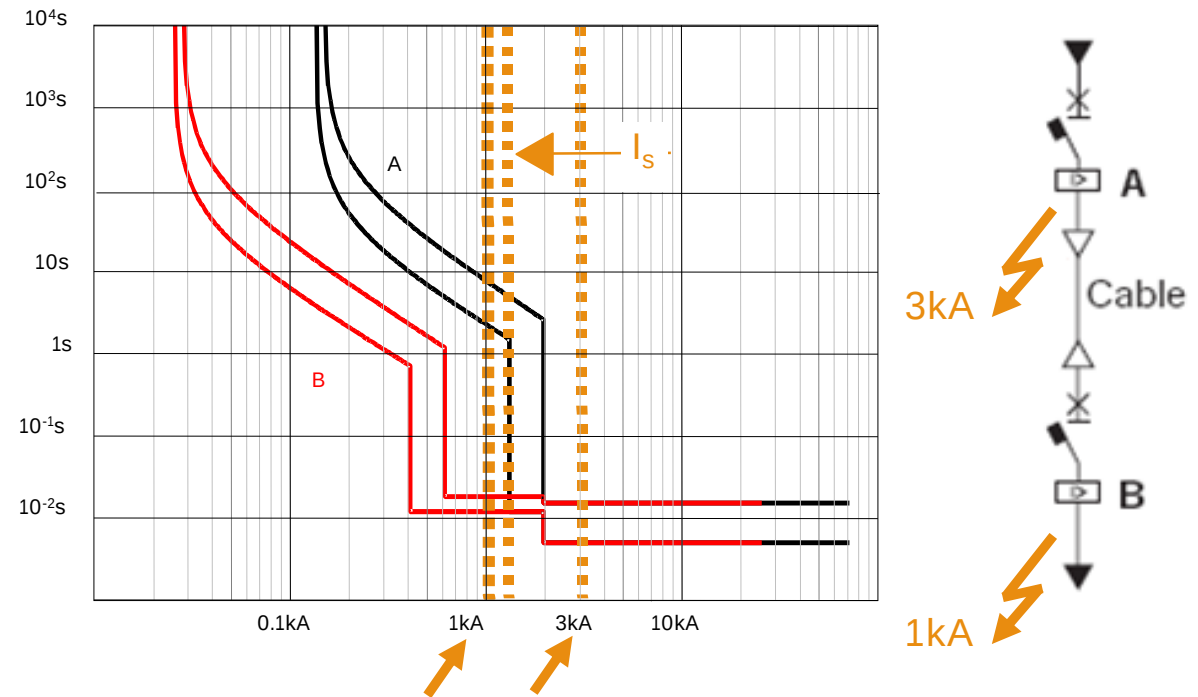
Example

Circuit breaker A will be set to a value which does not trip for faults which occur on the load side of B. ($I_{3Amin} > 1kA$)

Circuit breaker B will be set to trip for faults which occur on its load side ($I_{3Bmax} < 1kA$)

$$I_s = I_{3Amin}$$

Here the selectivity is a total selectivity, because it is guaranteed up to the maximum value of the short-circuit current, 1kA.

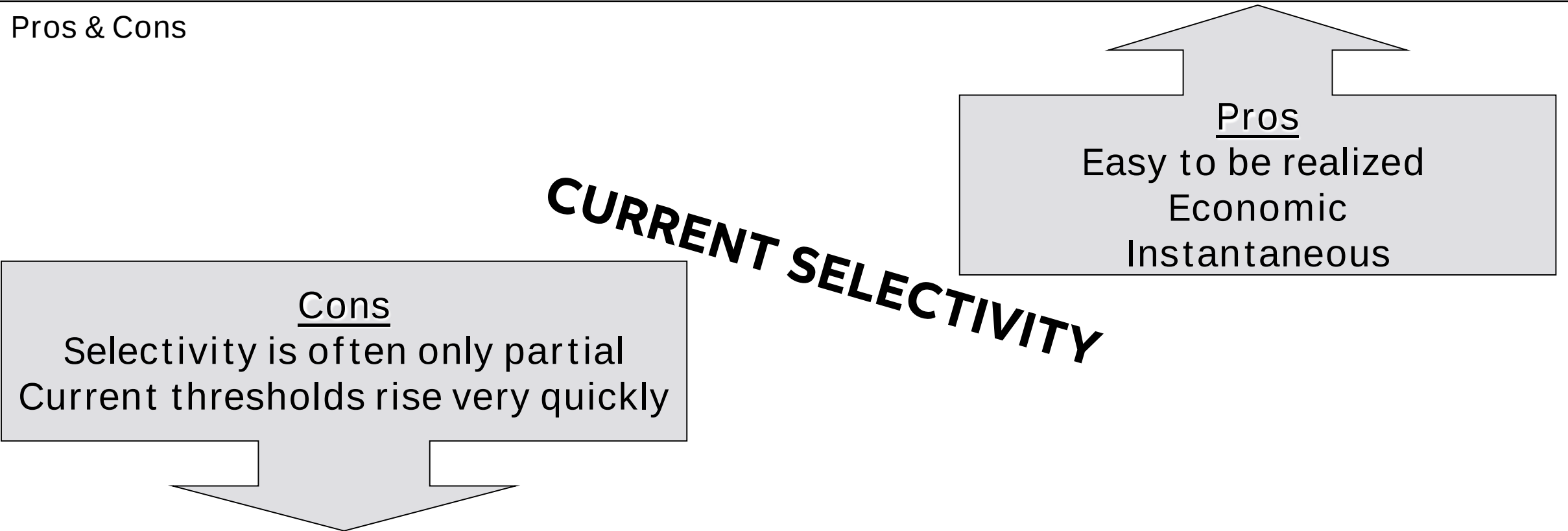


Low voltage selectivity

Selectivity techniques

Current selectivity

Pros & Cons



Low voltage selectivity

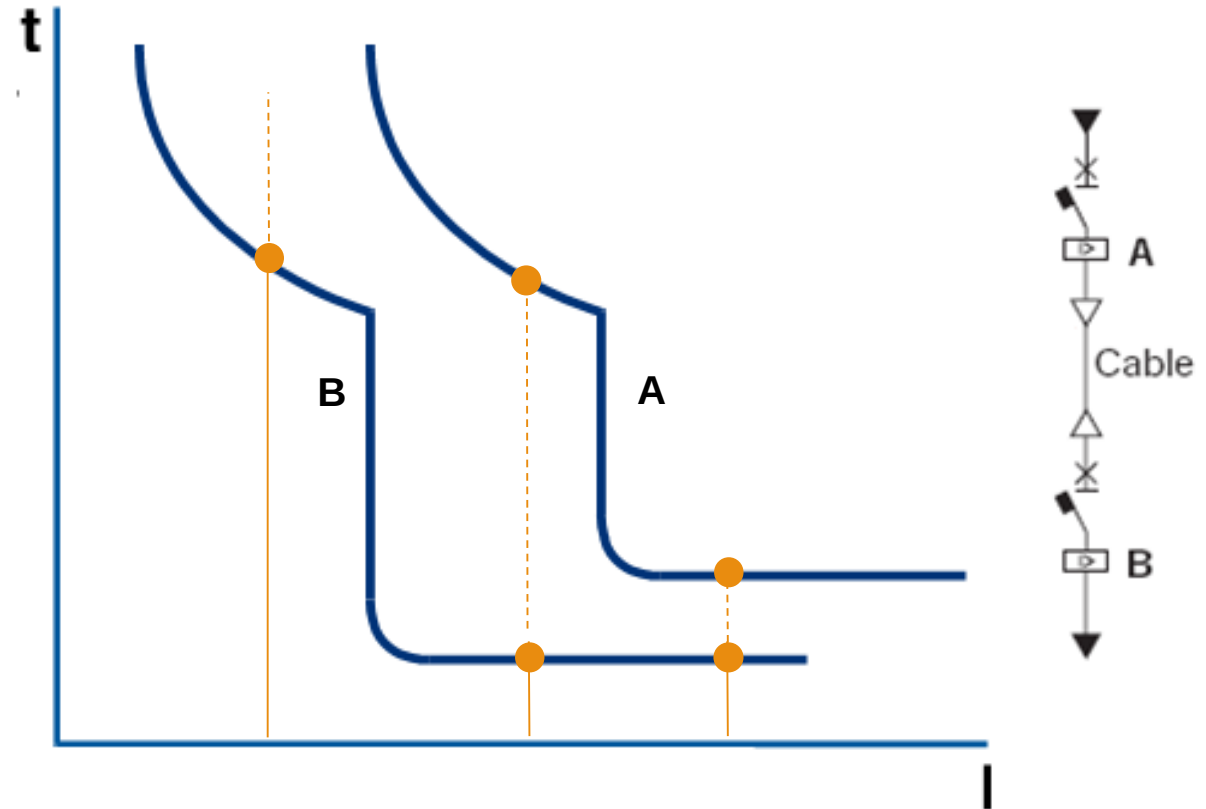
Selectivity techniques

Time selectivity

Basic concept

Time selectivity is based on a trip delay of the upstream circuit breaker, so to let to the downstream protection the time suitable to trip

- Setting strategy:
Progressively increase the trip delays getting closer to the power supply source
- On the supply side:
The S function is required



Low voltage selectivity

Selectivity techniques

Time selectivity

Example

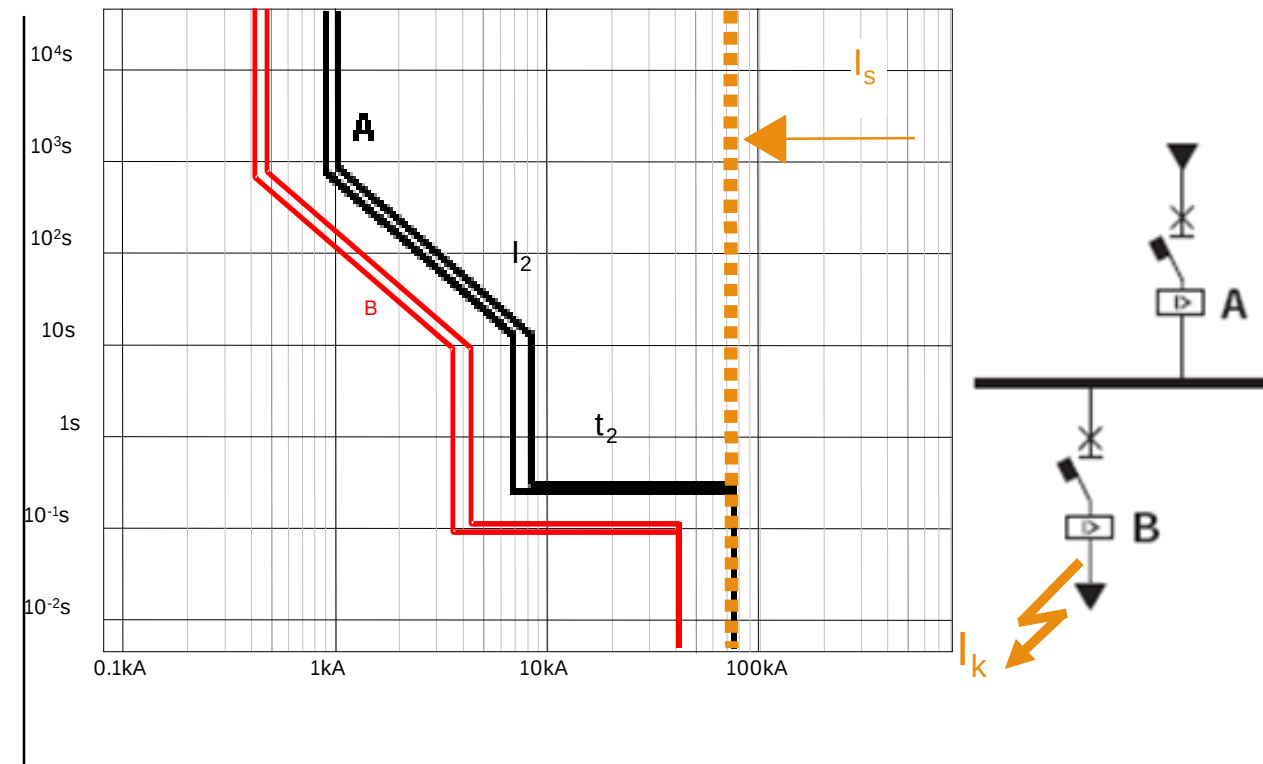
Circuit breaker A will be set with the current threshold I_2 adjusted so as to create an overlapping trip and with a trip time t_2 adjusted so that B always clears the fault before A.

B will be set with an instantaneous trip against short-circuit.

The ultimate selectivity value is:

$$I_s = I_{cWA} \quad (\text{if function I = OFF})$$

$$I_s = I_{3minA} \quad (\text{if function I = ON})$$



Low voltage selectivity

Selectivity techniques

Time selectivity

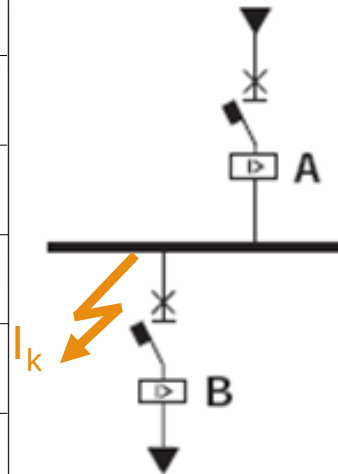
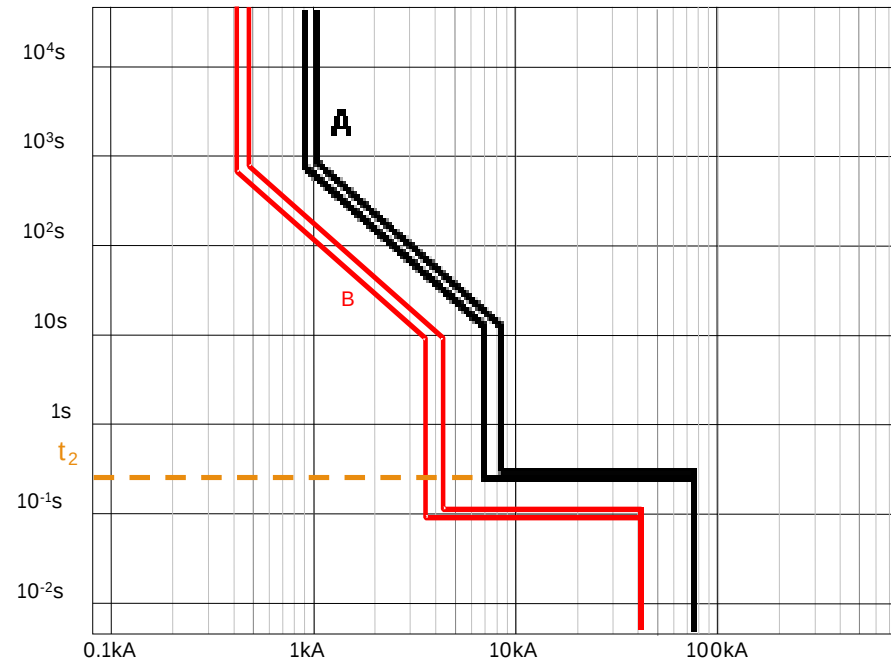
Example

There is a problem with time selectivity!

In the case of fault occurring at the busbars, circuit breaker A takes a delayed trip time t_2

The network must withstand high values of let-through energy!

If there are many hierarchical levels, the progressive delays could be significant!

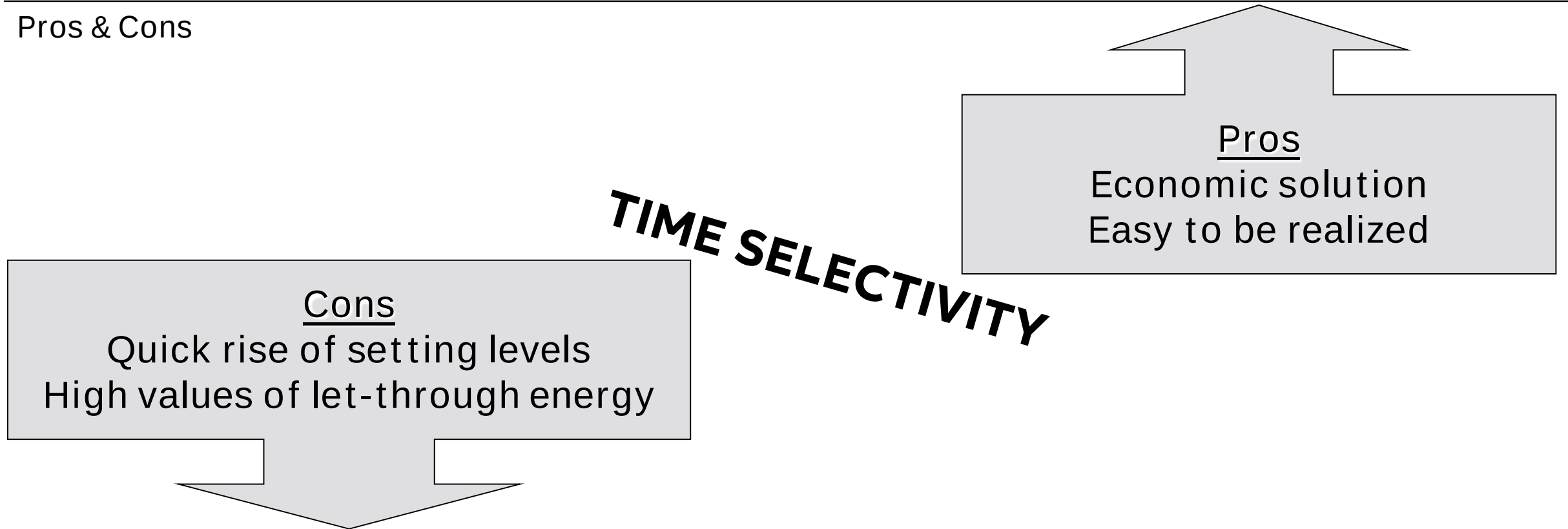


Low voltage selectivity

Selectivity techniques

Time selectivity

Pros & Cons



Low voltage selectivity

Selectivity techniques

Energy selectivity

Basic concept

Energy selectivity is based on the current-limiting characteristics of some circuit breakers

Current-limiting circuit breaker has an extremely fast trip time, short enough to prevent the current from reaching its peak

The ultimate current selectivity values are given by the manufacturer (Coordination tables)

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Power and productivity
for a better world™ **ABB**



Motor protection	Selectivity	Back-up	Other devices protection
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SOC - Selected Optimized Coordination

 Motor Protection Coordination tables for motor starting and protection.	 Selectivity Selectivity coordination tables between short circuit protection devices.	 Back-Up Back-up coordination tables between short-circuit protection devices.	 Other devices protection Coordination table for the protection of switch-disconnector and other devices by short circuit protection devices.
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Low voltage selectivity

Selectivity techniques

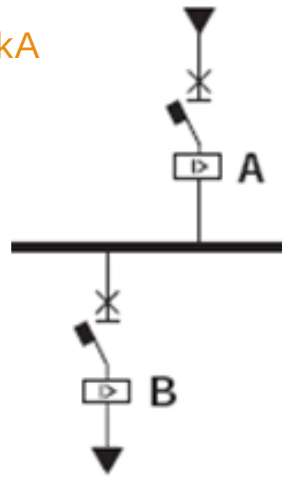
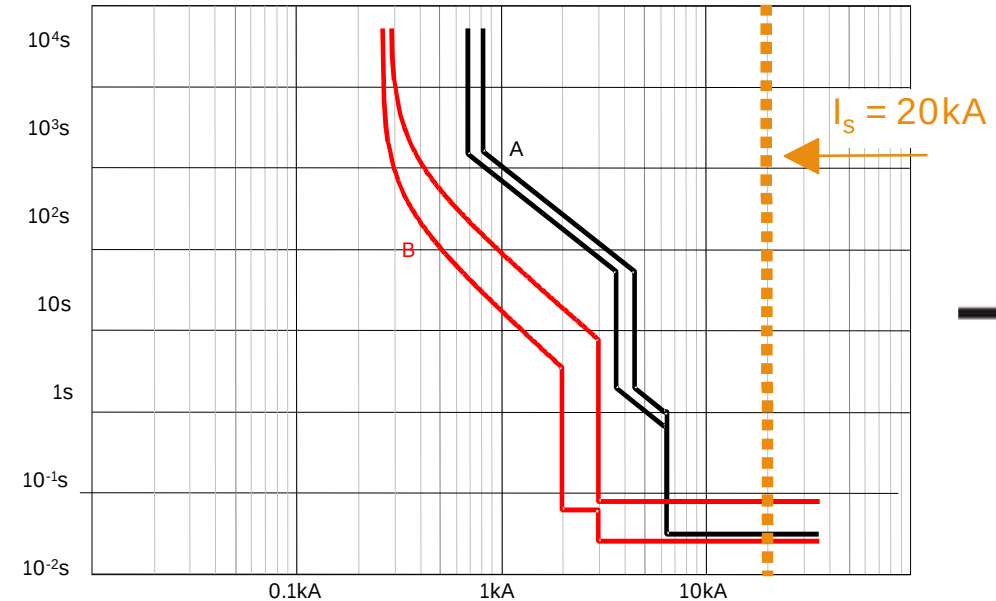
Energy selectivity

Example

Circuit breaker A settings:

I₃ = OFF

S protection is used for time selectivity

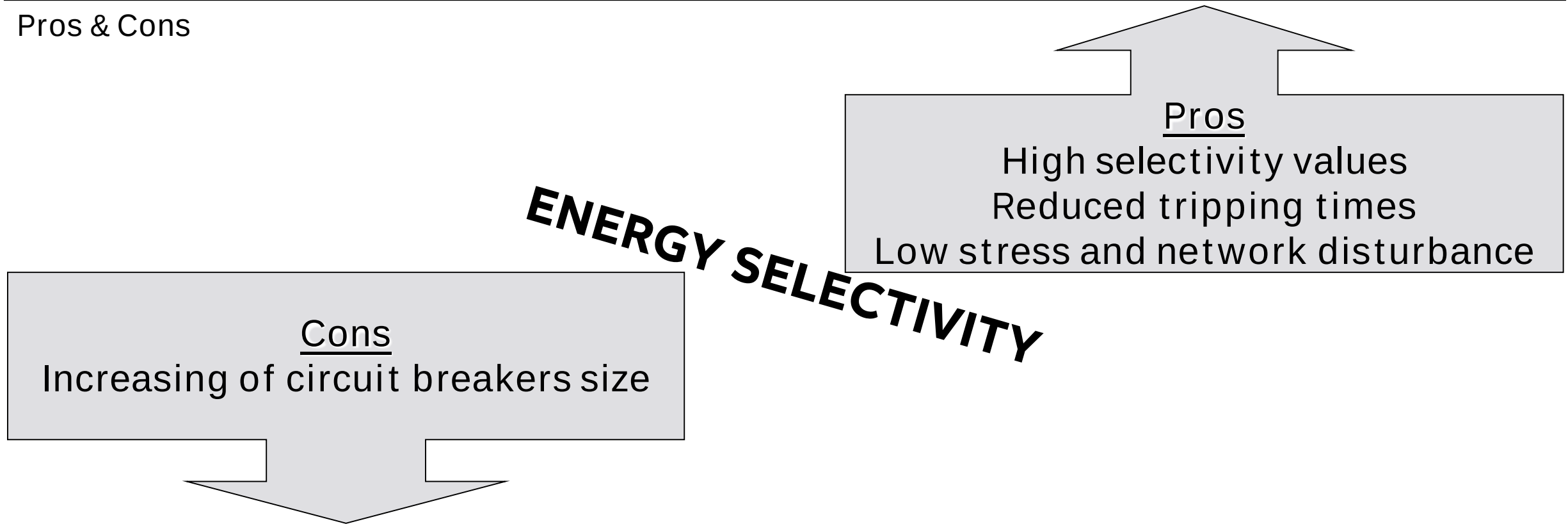


Low voltage selectivity

Selectivity techniques

Energy selectivity

Pros & Cons



Low voltage selectivity

Selectivity techniques

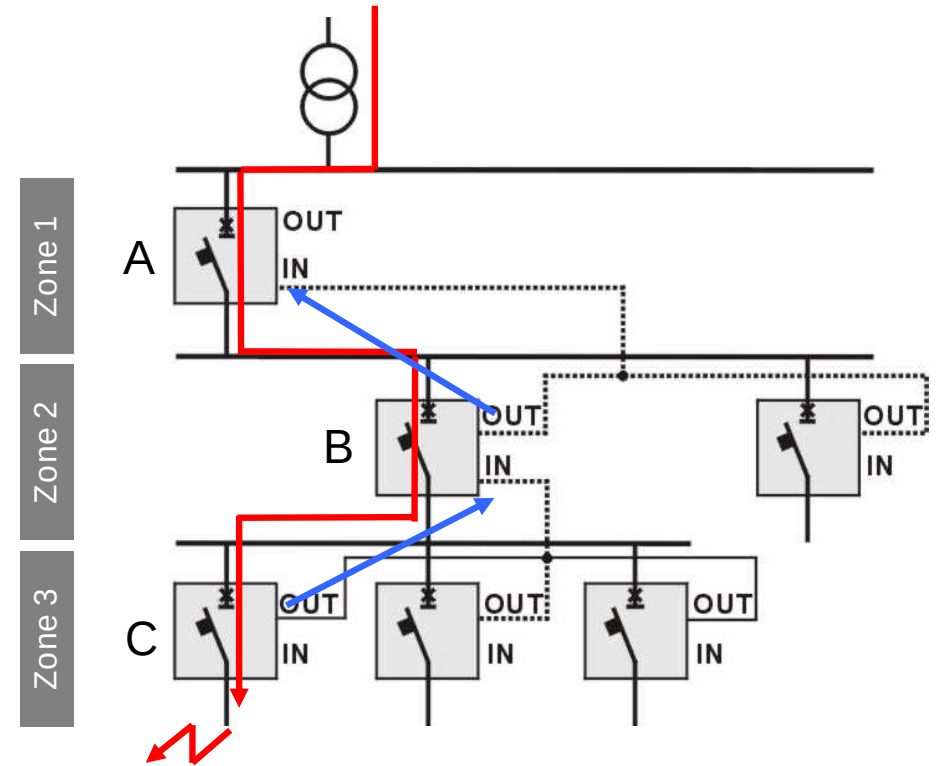
Zone selectivity

Example

A Does not open

B Does not open

C Opens

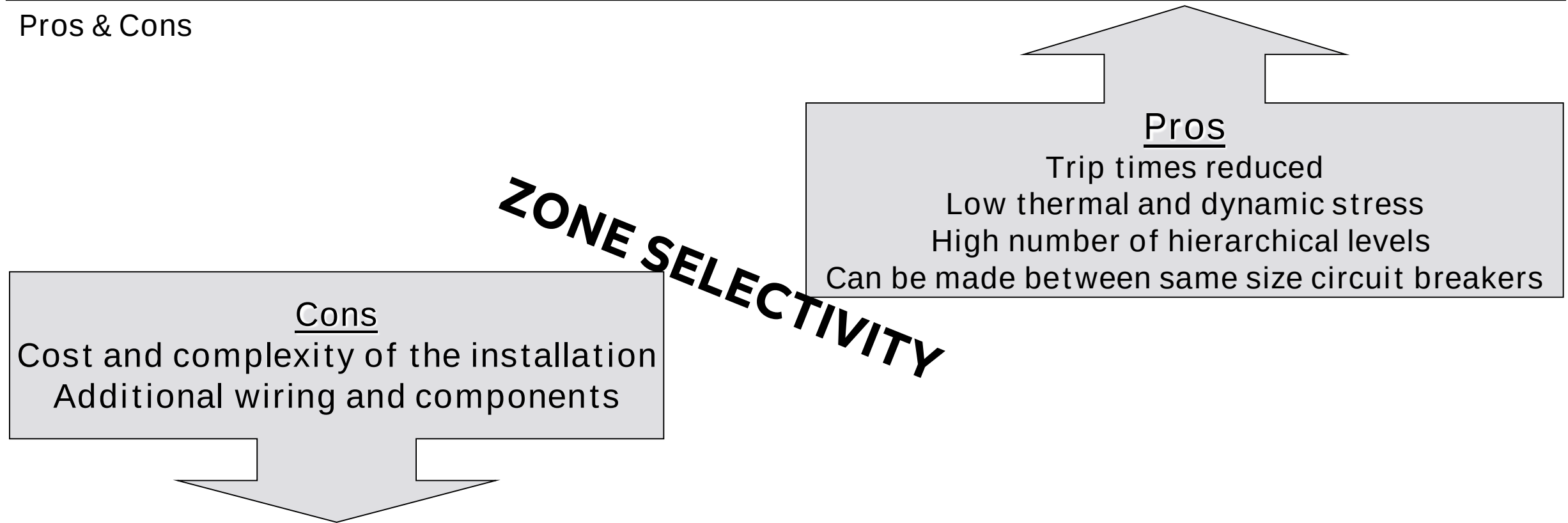


Low voltage selectivity

Selectivity techniques

Zone selectivity

Pros & Cons

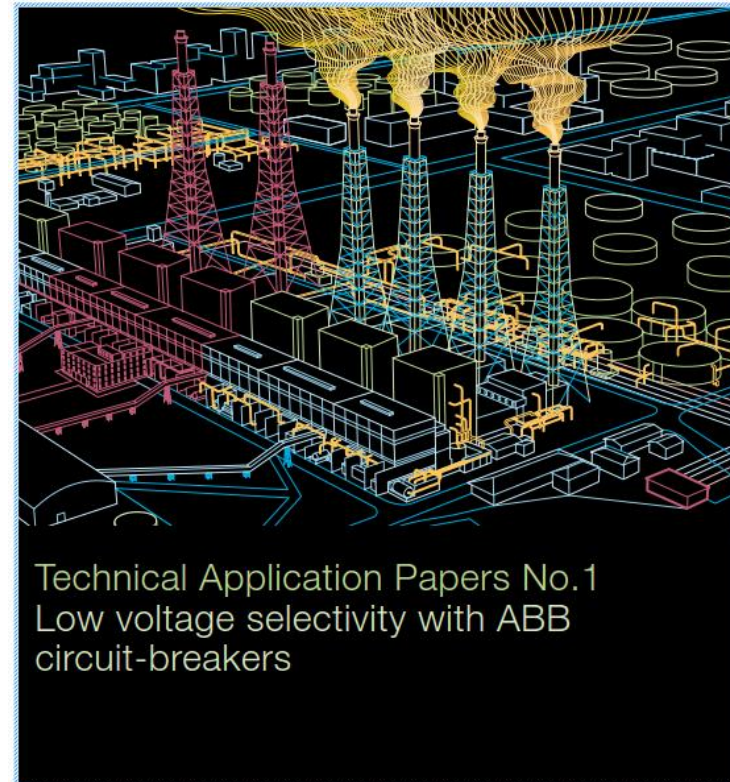


Low voltage selectivity

Selectivity definitions & standards

Agenda

- Definitions & Standards
- Selectivity Techniques
- Back-Up Protection



Low voltage selectivity

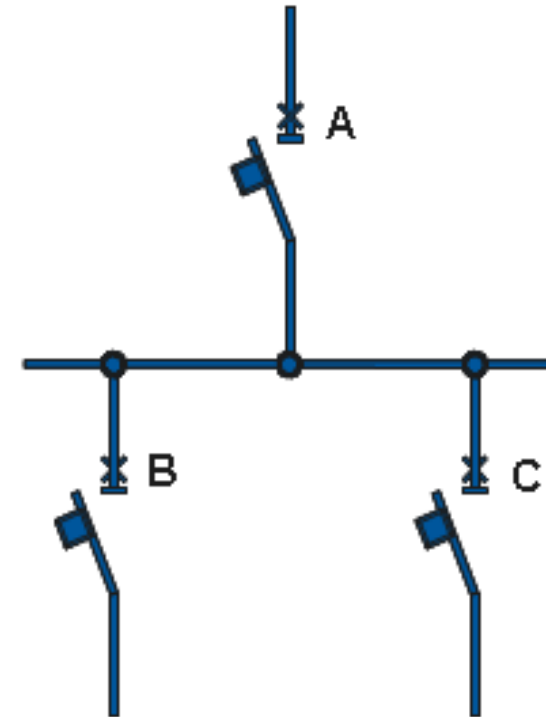
Back-Up Protection

What is Back-Up protection?

Back-Up protection (or Cascading)

Is a type of coordination of two protective devices in series which is done in electrical installations where continuous operation is not an essential requirement.

Back-up protection
excludes the use
of selectivity!!!



Low voltage selectivity

Back-Up protection

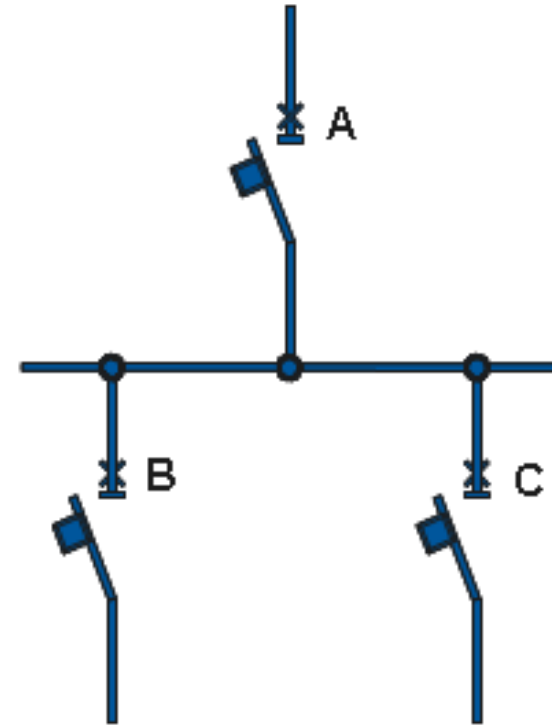
Back-Up protection

Basic concept

Back-up is used by those who need to contain the plant costs

The use of a current-limiting circuit breaker on the supply side permits the installation of lower performance circuit breakers on the load side

Both the continuity of service and the selectivity are sacrificed



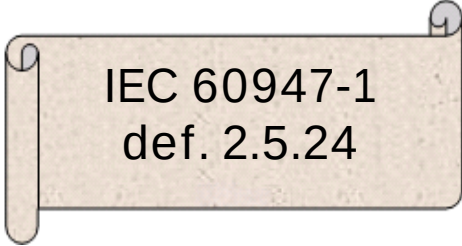
Low voltage selectivity

Back-Up protection

Standards definition

The definition of back-up is given by

IEC 60947-1 Standard: “Low voltage equipment
Part 1: General rules for low voltage equipment”



IEC 60947-1
def. 2.5.24

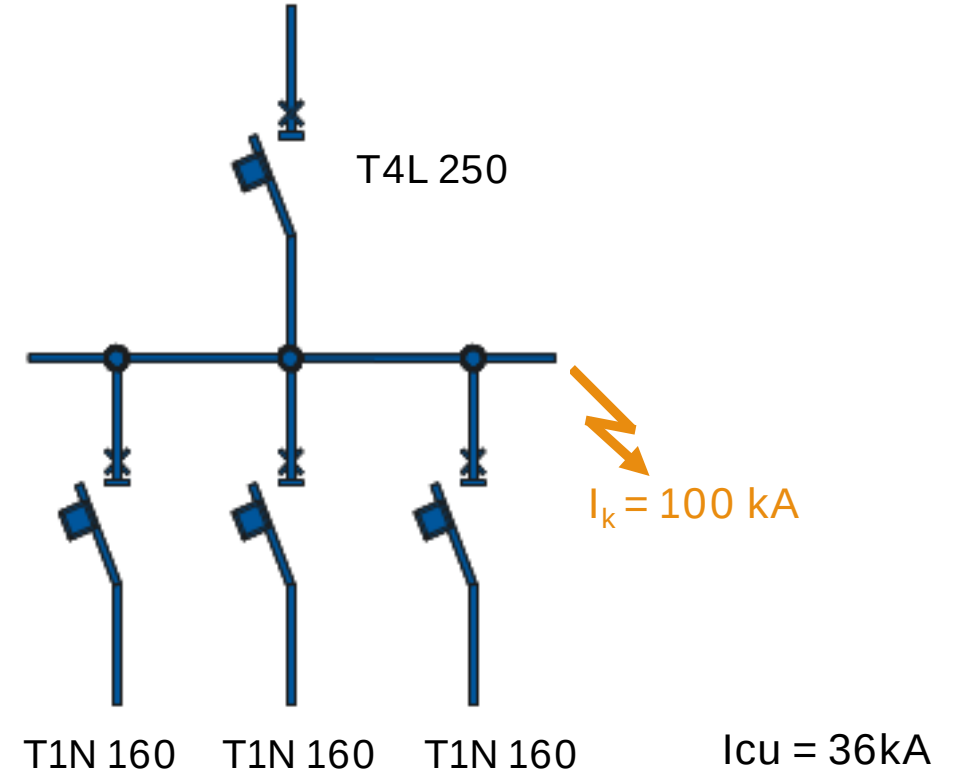
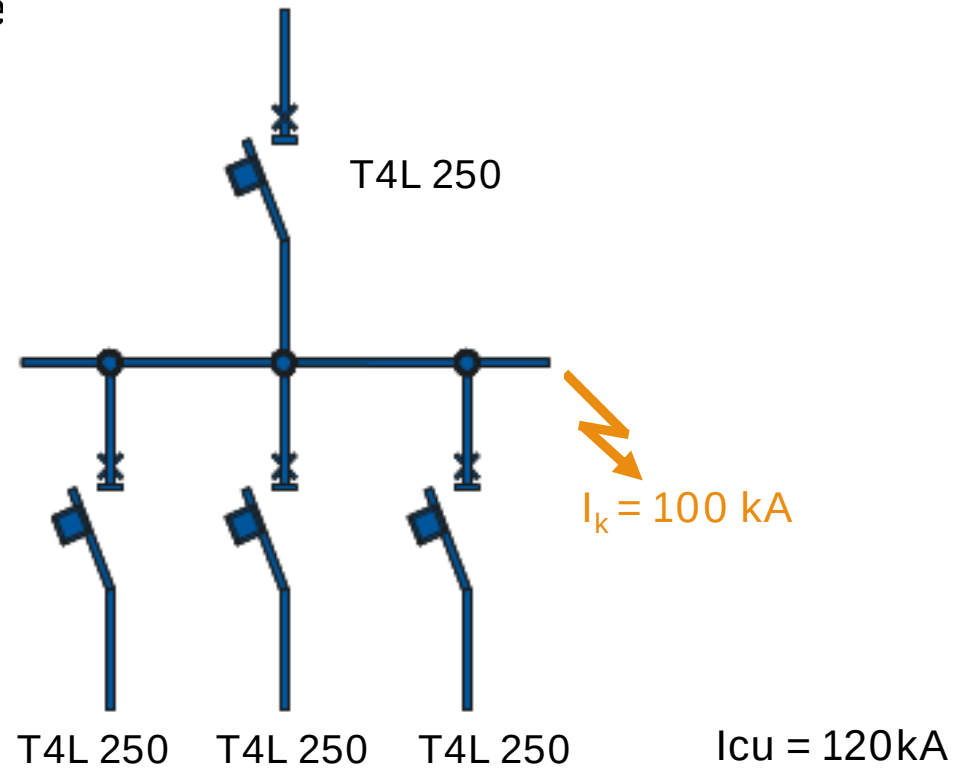
“Back-up is a coordination of two overcurrent protective devices in series, where the protective device on the supply side, with or without the assistance of the other protective device, trips first in order to prevent any excessive stress on downstream devices”.

Low voltage selectivity

Back-Up protection

Back-Up protection

Example



Low voltage selectivity

Back-Up protection

Back-Up protection table

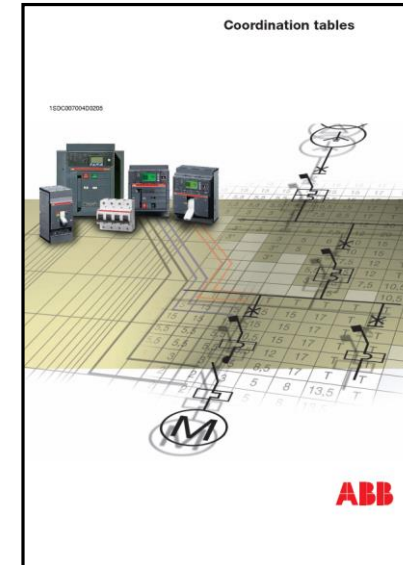
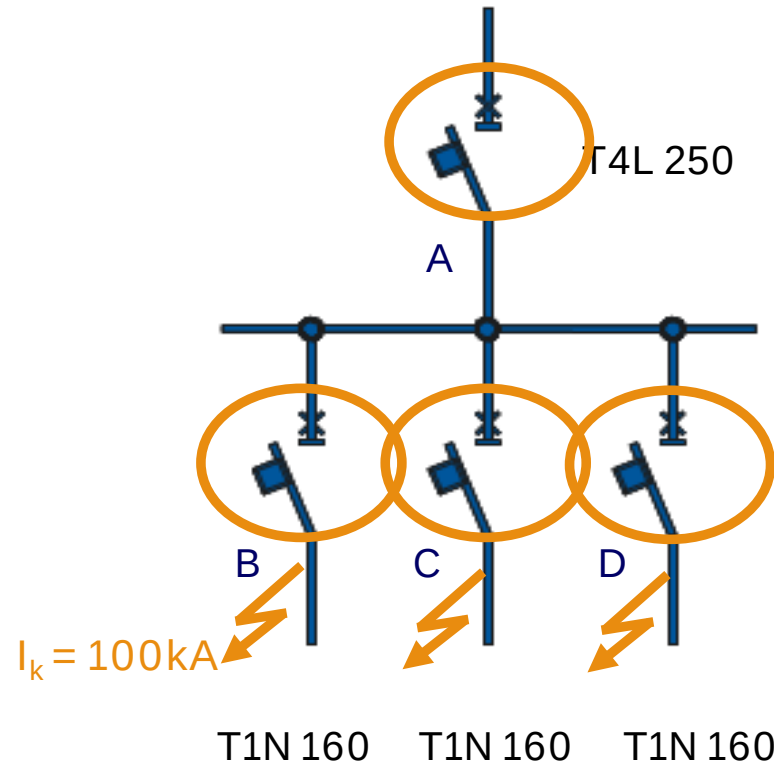
MCCB - MCCB @ 415 V																											
		Supply S.	T1	T1	T2	T3	T4	T5	T6	T2	T3	T4	T5	T6	T7	T2	T4	T5	T6	T7	T2	T4	T5	T6	T7	T4	T5
		Version	C	N						S						H					L	L	L	V			
Load S.	Char.	I _{cu} [kA]	25	36						50						65					85	120	100'	200			
T1	B	16	25	36	36	36	30	30	30	50	50	36	36	36		70	40	40	40		85	50	50	50		85	65
T1	C	25		36	36	36	36	36	36	50	50	40	40	50	50	70	65	65	65	50	85	85	85	70	50	130	100
T1	N	36								50	50	50	50	50	50	70	65	65	65	50	85	100	100	70	50	200	120
T2										50	50	50	50	50	50	70	65	65	65	65	85	100	100	85	85	200	120
T3											50	50	50	50	50		65	65	65	50		100	100	100	50	200	120
T4												50	50	50	50		65	65	65	50		100	100	65	65	200	120
T5													50	50	50			65	65	50			100	85	65		120
T6														50	40				65	40				70	50		
T2	S	50															70	70	70	70	85	100	100	85	85	200	130
T3																	70	70	70			100	100	100		200	150
T4																	70	70	70	70		100	100	85	85	200	150
T5																		70	70	70			100	85	85		150
T6																			70					85	85		
T2	H	70																		85	120	120	85	85	200	150	
T4																						120	120	100	100	200	180
T5																							120	100	100		180
T6																								100	85		
T2	L	85																				120	120			200	180
T4		120																								200	200
T5																										200	200

Low voltage selectivity

Back-Up protection

Back-Up protection

Example



$$I_{cu} (T4L+T1N) = 100\text{kA}$$

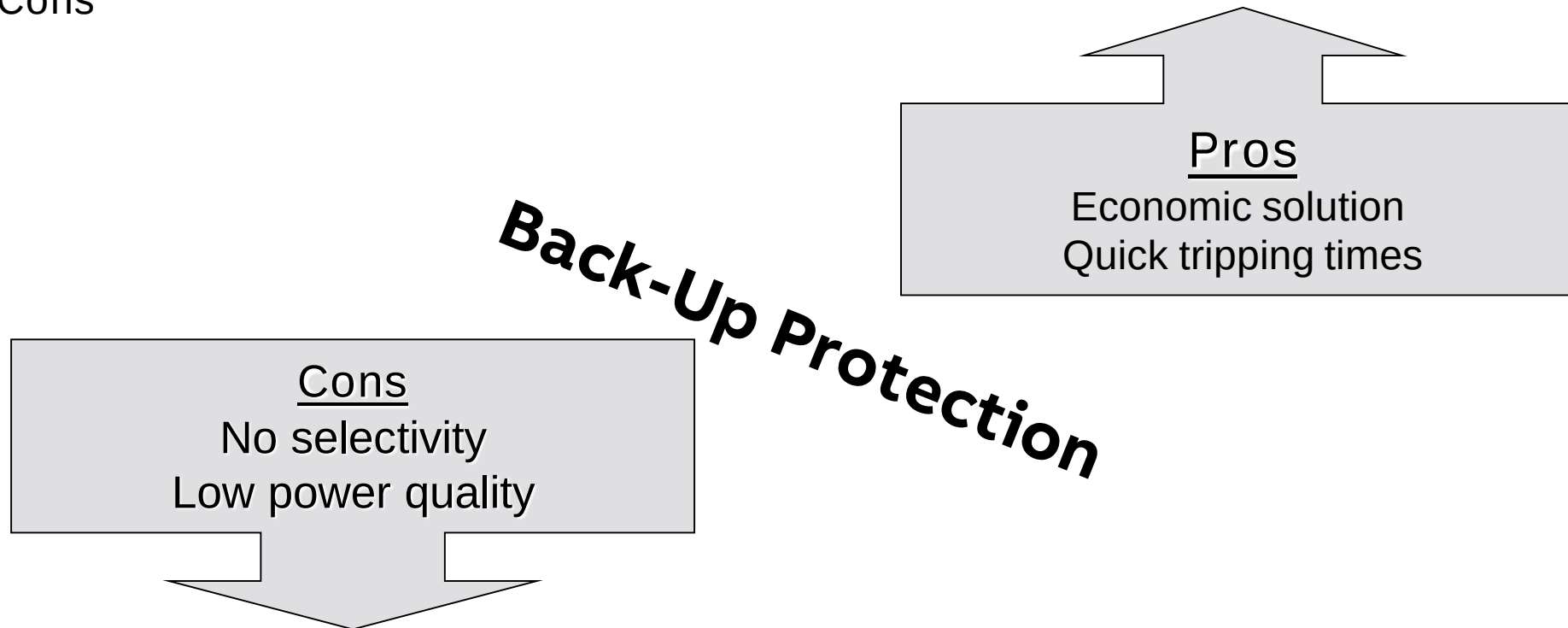
General power supply is always lost

Low voltage selectivity

Back-Up Protection

Back-Up protection

Pros & Cons



Low voltage selectivity

Selection of Protective & Control devices

Selected Optimized Coordination (SOC)

SOC - Select optimized coordination – online tool

Link: www.abb.com/abb.com/soc2

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Motor protection Selectivity Back-up Other Devices protection

SOC - Selected Optimized Coordination

			
Motor protection Coordination tables for motor starting and protection.	Selectivity Selectivity coordination tables between short circuit protection devices.	Back-up Back-up coordination tables between short-circuit protection devices.	Other Devices protection Coordination table for the protection of switch-disconnector and other devices by short circuit protection devices.



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